Heterogeneous Transfer Learning for Hyperspectral Image Classification Based on Convolutional Neural Network

Xin He, Yushi Chen, *Member, IEEE*, and Pedram Ghamisi, *Senior Member, IEEE*

***Abstract*— Deep convolutional neural networks (CNNs) have shown their outstanding performance in the hyperspectral image (HSI) classification. The success of CNN-based HSI classification relies on the availability sufficient training samples. However, the collection of training samples is expensive and time consuming. Besides, there are many pretrained models on large-scale data sets, which extract the general and discriminative features. The proper reusage of low-level and midlevel representations will significantly improve the HSI classification accuracy. The large-scale ImageNet data set has three channels, but HSI contains hundreds of channels. Therefore, there are several difficulties to simply adapt the pretrained models for the classification of HSIs. In this article, heterogeneous transfer learning for HSI classification is proposed. First, a mapping layer is used to handle the issue of having different numbers of channels. Then, the model architectures and weights of the CNN trained on the ImageNet data sets are used to initialize the model and weights of the HSI classification network. Finally, a well-designed neural network is used to perform the HSI classification task. Furthermore, attention mechanism is used to adjust the feature maps due to the difference between the heterogeneous data sets. Moreover, controlled random sampling is used as another training sample selection method to test the effectiveness of the proposed methods. Experimental results on four popular hyperspectral data sets with two training sample selection strategies show that the transferred CNN obtains better classification accuracy than that of state-of-the-art methods. In addition, the idea of heterogeneous transfer learning may open a new window for further research.**

***Index Terms*— Classification, convolutional neural network (CNN), hyperspectral image (HSI), transfer learning.**

I. Introduction

HYPERSPECTRAL sensors obtain the images of a target across a wide range of electromagnetic spectrum. Therefore, hyperspectral image (HSI) contains abundant spectral and spatial information simultaneously. Due to the unique characteristics, HSI has been widely used in many areas, including agriculture, medicine, and remote sensing [1]. On the other hand, the processing of HSIs is a challenge due to the fact that HSIs are quite different from other optical data sets (e.g., RGB and multispectral images) [2].

In the last few decades, an impressive number of HSI classification methods have been proposed. In this context, many supervised learning methods in machine learning community have been investigated and modified for HSI classification [3], [4].

In general, there are two types of HSI classification methods: spectral classifiers and spectral–spatial classifiers [5]. For spectral classifiers, only spectral information obtained from a hyperspectral sensor is used for classification [6]. On the other hand, compared with spectral classification, spectral–spatial classifiers incorporate the spatial information around the pixel to be classified [7].

Early HSI classification methods mainly focused on dimensionality reduction, which includes band selection and feature extraction, followed by classification. Band selection methods select the most relevant bands to mitigate the curse of dimensionality [8]. Many information-theoretic criteria have been suggested for optimal band selection, such as mutual information [9] and manifold ranking [10]. Meanwhile, dimensionality reduction of HSI included principal component analysis (PCA) [11], independent component analysis [12], and linear discriminant analysis [13]. Among the spectral classifiers, we can mention support vector machines (SVMs) [3], k-nearest neighbor approach [14], multinomial logistic regression [15], decision trees [16], random forest [17], and AdaBoost [18].

Spectral–spatial classifiers integrate both spectral and rich spatial features to boost classification performance. For example, Fauvel *et al.* [19] fused original HSI and morphological information to exploit the spectral–spatial information via multiple morphological operations. In [20], a novel fusion and classification method was developed via extinction profiles for HSI classification. In [21], edge-preserving filtering was proposed to optimize pixel-wise classification maps in a local filtering framework. Other techniques exploited spectral–spatial information based on Markov random fields to boost the classification accuracy [22], [23]. In addition, varieties of kernel learning methods are widely applied to spectral–spatial feature extraction. The core concept of these methods is to map the data from the original feature space to a convenient feature subspace [24]–[27]. The traditional research works for HSI classification have achieved satisfactory performance, but they

Manuscript received July 7, 2019; revised October 10, 2019; accepted October 30, 2019. Date of publication November 26, 2019; date of current version April 22, 2020. This work was supported by the Natural Science Foundation of China under Grant 61971164. *(Corresponding author: Yushi Chen.)*

X. He and Y. Chen are with the School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin 150001, China (e-mail: 19b905007@stu.hit.edu.cn; chenyushi@hit.edu.cn).

P. Ghamisi is with the Helmholtz-Zentrum Dresden-Rossendorf, Helmholtz Institute Freiberg for Resource Technology, 09599 Freiberg, Germany (e-mail: p.ghamisi@gmail.com).

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2019.2951445

0196-2892 © 2019 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

heavily rely on handcrafted descriptors, and the corresponding parameter setting always depend on the expert knowledge. To further enhance the classification performance of HSI, the development of more powerful models is necessary.

Deep learning methods have become a hot topic, since AlexNet won the ImageNet classification challenge in 2012. AlexNet can extract discriminative features in an end-to-end manner. Recently, numerous deep learning-based methods have been introduced for HSI classification. Chen *et al.* [28] proposed a deep feature extraction method based on stacked autoencoder (SAE) for HSI classification, which combines PCA-based dimensionality reduction and logistic regression classification. In [29], a deep belief network (DBN) was investigated to learn the restricted Boltzmann machine network layer by layer. However, the two methods need many parameters to train in networks, which are composed of fully connected (FC) layers. In comparison with SAE and DBN, a convolutional neural network (CNN) has the characteristics of local connections and weight sharing, which can decrease numbers of parameters [30].

Numerous CNN-based approaches have been developed to extract and utilize spectral–spatial features for HSI classification and representation [31]. Yue *et al.* [32] proposed a deep CNN architecture to extract high-level spatial and spectral information for HSI classification. Chen *et al.* [33] presented a regularized 3-D CNN network to extract effective features for HSI spectral–spatial classification. Moreover, the CNN models can be integrated with other spatial feature extractions (i.e., morphological profiles and Gabor filtering) to enhance the final performance [34], [35]. In [36], a novel strategy was designed with pixel pairs that were classified by the trained CNN, for which a voting strategy was used to obtain the final classification map. In [37], the balanced local discriminant embedding algorithm was proposed for spectral feature extraction, which merged spatial features extracted by a CNN. However, the abovementioned methods combine different kinds of techniques with CNN without fully utilizing spectral–spatial correlation information. Instead of calculating 2-D convolution kernels for each pixel of HSIs, the 3-D convolutional kernels can extract spectral–spatial features for pixel-wise HSI classification [38], [39]. Besides, data augmentation and network optimization approaches based on CNN have been proposed to deal with limited training samples in the HSI classification [40]–[42].

Furthermore, some advanced deep CNN schemes, including deep residual networks and densely connected CNN, have been introduced to extract highly discriminative spectral–spatial features for HSIs, which contain shortcut connections between the layers. The generalization ability of the network can be enhanced with the combination of various spectral–spatial features across the layers [43], [44].

The abovementioned articles have focused on improving classification accuracy by developing a range of advanced methods, which adopt random sampling strategy. Apart from the traditional random sampling strategy, recently, some researchers have studied novel sampling strategies. For example, [45] and [46] presented a more comprehensive study on different strategies for the selection of training samples, called the controlled random sampling strategy, to reduce the overlap between the training and test samples. Similar to the abovementioned works, novel sampling strategies have been investigated based on cluster sampling [47], [48].

Although deep-learning-based methods have achieved outstanding performance for HSI classification, there are still many difficulties in practice. First, due to the fact that deep models usually contain a large number of trainable parameters (i.e., weights), it is difficult to train the models. Second, in order to properly train the parameters, many training samples are needed in deep-learning-based methods. Unfortunately, limited training samples are a common issue in remote sensing applications. Third, the back-propagation-based training methods of deep models are time-consuming.

In order to address the aforementioned problems, transfer learning is investigated in this article. Transfer learning, which is a technique aiming at transferring the knowledge between two or more related domains [49]. In this article, the VGGNet (very deep convolutional networks) [50] is applied, which won the second place in the classification task of "ImageNet Large Scale Visual Recognition Challenge (ILSVRC)" in 2014 [51]. The proper usage of transfer learning will considerably reduce the number of training samples and shorten the training time simultaneously.

The optical images and HSI are quite different. Therefore, in order to harness the powerfulness of transfer learning and deep learning, a heterogeneous transfer learning based on CNN (HT-CNN) is proposed for HSI classification.

Furthermore, in order to mitigate the negative influence caused by the difference of optical data set and HSI, attention mechanism is used in this article. In the fine-tuning step of transferring the CNN, feature maps obtained in the pretrained model are reweighted, which further improve the classification performance of HT-CNN.

Finally, two different training sample selection strategies are investigated in this article to validate the proposed methods mentioned earlier.

The main contributions of this article are summarized as follows.

1) A new method, HT-CNN, is proposed for HSI classification. The method is a combination of a transformation between two heterogeneous data sets, transfer learning, and CNN.
2) In order to mitigate the difference of the two data sets, the attention mechanism is used to reweight the feature maps, which further improves the classification accuracy of HSI.
3) Centered kernel alignment (CKA) is introduced to evaluate the similarity between the pretrained model and the fine-tuned model.
4) Four widely used hyperspectral data sets and two training sample selection strategies (i.e., random sampling and controlled random sampling) are used to provide a comprehensive evaluation of the proposed classification methods.

The rest of this article is organized into four sections. Section II describes the proposed heterogeneous transferring CNN for HSI classification, and Section III presents the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

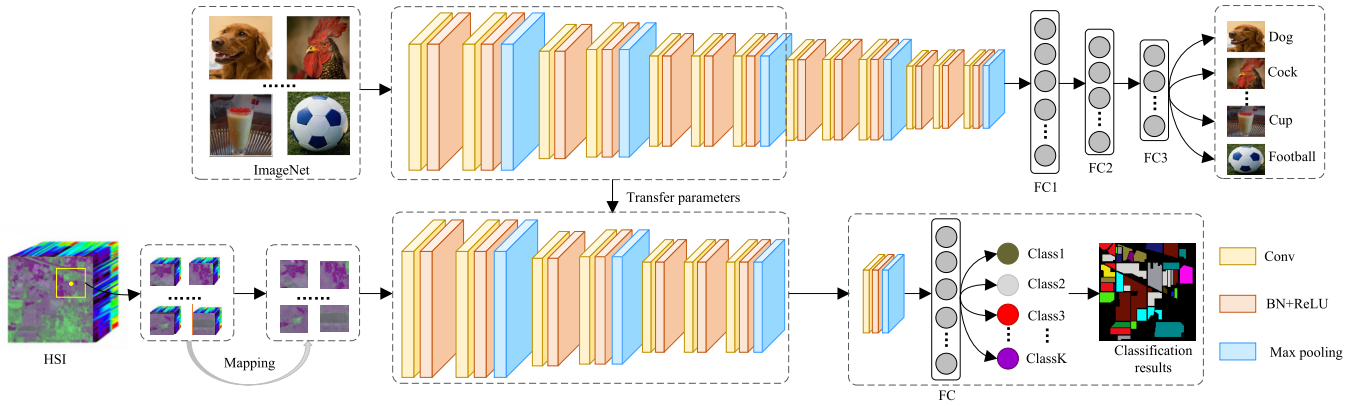


Fig. 1. Framework of heterogeneous transfer learning for HSI classification.

HT-CNN with attention classification method. Experimental results are shown in Section IV. Section V concludes the observations and completes this article.

II. Heterogeneous Transferring CNN for HSI Classification

The framework of the proposed method is shown in Fig. 1. First, a mapping layer is used to handle the issue of different numbers of channels between the original data set (i.e., ImageNet) and the target data set (i.e., HSI). Then, the first several layers of the pretrained CNN (i.e., VGGNet) are transferred to the target CNN model. After that, a well-designed CNN with an FC layer is used to perform the classification task.

A. Mapping From Heterogeneous Data Sets

CNN, which is a method loosely inspired by neuroscience, tries to extract hierarchical features of inputs [52]. The first few layers of visual systems and CNNs extract the low-level features, such as blobs, corners, and edges, which are the generic image representations [49]. Therefore, the CNN trained on one source data set (e.g., ImageNet) can be reused on other target data sets (e.g., HSI).

ImageNet is a benchmark dataset, which contains hundreds of classes and millions of images. In the recent years, a huge number of well-designed CNNs, including AlexNet [30], VGGNet [50], GoogleNet [53], and ResNet [54], have been trained on the ImageNet dataset. The first few layers of the pretrained CNN models can be reused on HSIs. Unfortunately, there are some difficulties in using the pretrained CNN for HSI. For example, the images of the ImageNet dataset contain three channels (i.e., R, G, and B) but HSI usually contains hundreds of bands. The first step of heterogeneous transfer learning is to build a mapping layer and translate the HSI to a dataset with three channels.

Let $O \in R^{W\times H\times B}$ be the input HSI, in which $W \times H$ represents the spatial dimensions, B represents the total number of spectral bands, $O' \in R^{W\times H\times 3}$ is the transformed data set, and $T \in R^{B\times 3}$. Therefore

$$O' = O \times T. \tag{1}$$

There are $B \times 3$ learnable parameters in the transformation. The mapping layer is combined with other layers to form an end-to-end learning system. Fig. 2 shows the process of mapping.

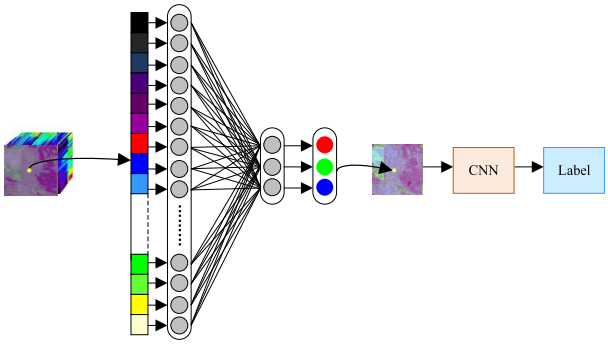


Fig. 2. Illustration of mapping from the heterogeneous data sets.

All the parameters, including the parameters in the mapping layer, are trained through backpropagation with the help of training samples.

B. CNN for HSI Classification

The classical CNN consists of multiple layers, including the convolutional layer, nonlinear transform layer [e.g., rectified linear function (ReLU)], and pooling layer. The main element of CNN is the convolutional layer, which outputs feature maps by computing a dot product between the receptive fields. A convolutional layer is calculated by

$$h_{i,j} = f\left(\sum_{m}^{M}\sum_{n}^{N} w_{m,n}x_{i+m,j+n} + b_{m,n}\right) \tag{2}$$

where x denotes the input, w denotes the shared parameters, b and h represent bias and the output neuron, respectively, and m, n, i, and j are the parameter indexes. Then, the nonlinear activation function (e.g., ReLU) is applied to the generated feature maps.

The most commonly used pooling operation is max pooling, which performs the downsampling operation via computing the maximum on a local region of a features' map. The FC layers follow the abovementioned layers, and the last FC layer, which is called softmax layer, computes values for each class.

Overfitting always exists in the training process of CNN-based methods. To overcome this issue, dropout is adopted in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

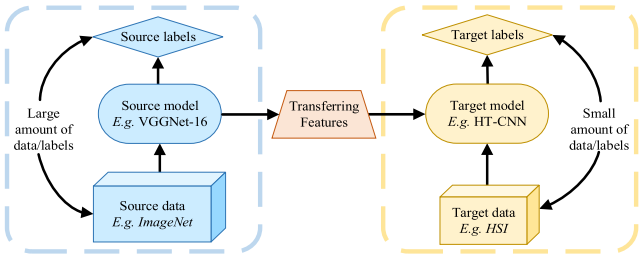


Fig. 3. Illustration of transfer learning.

the CNN models, which drops out features randomly [55]. The dropout operation prevents interdependences from emerging between the nodes. Besides, other regularization techniques also turn out to be significantly beneficial (i.e., global average pooling and $L2$ regularization), which alleviate the risk of overfitting problem in the process of learning network parameters [40], [42].

In addition, to enhance the performance of the networks, batch normalization (BN) is used, which computes mean and variance for a mini-batch version [56]. The CNN with BN not only learns faster but also increases the accuracy.

Based on the conventional CNN models, VGGNet is one of the influential CNN models, which considers the depth of appropriate layers and does not increase the total number of parameters in deep CNN models. The VGGNet addresses the depth of conventional architecture design. In this article, we mainly adopt VGGNet-16. It includes 13 convolutional layers with small 3×3 filters and three FC layers. Each convolutional layer is followed by the BN layer and ReLU operation. In the feedforward process, the model transforms the input from the original pixel values to the final class values. The whole parameters are trained through the backpropagation algorithm.

C. Transfer Learning With CNN

Transfer learning aims at extracting knowledge from the source data and applies it to the target data [57]. The core idea of transfer learning is shown in Fig. 3. The source task usually has a large number of training samples, while the target task usually has limited training samples. In transfer learning, the learned features from the source task can be used to improve the performance of the target task. For example, in neural network-based methods, the learned weights on the source data can be utilized to initialize the network of the target task. Therefore, if it is properly used, transfer learning can improve the performance of the target task when the number of training samples of the target task is limited. Hence, the transfer learning is explored for the HSI classification task.

CNN has the ability of learning discriminant and invariant representations with sufficient training samples, but it is expensive to obtain adequate amount of training data, which causes a serious overfitting problem. The CNN models based on the transfer learning strategy are one solution to solve the problem, which can obtain a better classification result with limited labeled samples.

The prior studies have proven that the learned feature representations in CNNs in the original task can be transferred to the new task [58]. Due to the similarity tasks between the ImageNet classification and the HSI classification, the transfer learning step can be facilitated by fine-tuning ImageNet on HSIs. In this article, VGGNet-16 is used as source network for the HSI classification task to achieve high classification accuracy.

In this article, the proposed framework for HSI classification with transfer learning is shown in Fig. 1. We fine-tune the weights of the pretrained network (VGGNet-16) by the backpropagation algorithm. In the design of the proposed network, the weights of the first seven layers are transferred to the weights of the target model. Since the first layers usually extract low-level features (i.e., blobs, corners, and edges), which can be transferred to the target classification task. However, the top layers of VGGNet-16 capture high-level features, which are more specific for the dedicated task. The detailed discussion of the transfer layer is shown in Section IV.

D. CKA

In order to deeply understand the transfer learning step, we utilize the CKA, which has been recently introduced as a similarity index. CKA describes the correspondence between the networks with different initialized settings. In this article, it is utilized to evaluate the similarity of learned features between the networks with and without transfer [59].

The CKA compares the similarity structures composing of the results of each pair of examples in each representation separately. It is composed of three steps. The first is to compute the similarity between the features by inner product, which can be defined as $\|Y^T X\|_F^2$, and $X \in R^{n \times p_1}$ and $Y \in R^{n \times p_2}$ are the matrices of p_1 or p_2 neurons for n samples, respectively. F represents the Frobenius norm. $Y^T X$ denotes the dot products between features of the ith and jth examples from X and Y, respectively, which describes the similarity relationship between the features of these examples belonging to their respective networks. In this article, X is the features from different convolutional layer in the network without fine-tuning, and Y is the features from the corresponding convolutional layer with fine-tuning. The examples are the randomly selected test samples. Every pair example consists of the learned features from different networks, and the final similarity results sum the squared dot products between every pair. Next step, the Hilbert–Schmidt independence criterion (HSIC) is utilized to reproduce the inner products from kernel Hilbert spaces [60]. Finally, the CKA is formed as follows:

$$\mathrm{CKA}(K, L) = \frac{\mathrm{HSIC}(K, L)}{\sqrt{\mathrm{HSIC}(K, K)\mathrm{HSIC}(K, L)}} \tag{3}$$

where K_{ij} and L_{ij} are computed by two kernels of $k(x_i, x_j)$ and $l(y_i, y_j)$ in the second step, respectively.

The abovementioned formulas describe the CKA with a linear kernel. Additionally, the CKA with the radial basis function (RBF) kernel can also be defined by bandwidth σ, which is defined as the fraction of the median distance between examples, which is shown as follows:

$$k(x_i, x_j) = \exp\left(-\|x_i - x_j\|_2^2/(2\sigma^2)\right). \tag{4}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

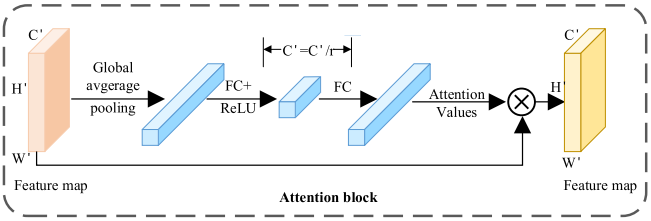


Fig. 4. Process of attention block.

III. Heterogeneous Transferring CNN With Attention for HSI Classification

The original data set and the task data set are quite different. Therefore, the feature maps of the heterogeneous data sets may quite different too. In order to further mitigate the difficulty of transfer learning between heterogeneous data sets for better classification accuracy, attention mechanism is used to adjust the transferred feature maps.

A. Attention Mechanism

Here, attention mechanism is introduced to adjust the feature maps of the pretrained model for better classification on the target data set.

The attention mechanism imitates the human biological system, which pays attention to relevant elements of the input and ignores other irrelevant information [61]. For example, when we look at a picture, although we can see the whole picture, our eyes only focus on a small piece carefully, which means that the attention system of human brains is not balanced and there exists the weight distinction. The core idea of attention mechanism is to give different weights to different information. Attention mechanism is used successfully in a series of works, including text entailment, sentence representations, and so on [62], [63].

Inspired by recent works in computer vision, we consider the usage of the attention mechanism to mitigate the difference between heterogeneous data sets. Specially, in this article, the importance of feature maps in low- and mid-level representations are obtained based on attention mechanism.

B. Feature Map Attention

Based on the attention mechanism to reduce the difference between heterogeneous data sets, the attention block is presented to express the importance of different feature maps according to their interdependences in each convolution layer. The following steps describe the attention block step-by-step (see Fig. 4).

First, the global average pooling operation is utilized here to capture the global information of every feature map in each channel. Every feature map is squeezed to a value by this operation, which can reflect the whole spatial information in each feature map, and it can be described as follows:

$$v = \frac{1}{H' \times W'} \sum_{q=1}^{H'} \sum_{s=1}^{W'} u(q, s) \tag{5}$$

where v is the output value and u is the feature map with the size o $f H' \times W'$.

The next step is to adjust the values of different feature maps based on their importance. In order to evaluate the importance of different feature maps, a neural network, which contains two FC layers and a ReLU operation, is carefully designed. The first FC layer is used to squeeze the total number of feature maps to reduce the number of parameters and, consequently, the computational complexity of the network. The size of the second FC layer aims to revert to the number of original feature maps. The ReLU operation is used to obtain the nonlinear relationships between different feature maps.

Finally, the output of the last step is scaled between 0 and 1 by the sigmoid function, which is called attention values for short. The attention values are the final results to reflect the importance of different feature maps. Next, the attention values are multiplied to each feature map. All the abovementioned steps are composed of the final attention block.

The attention block can be inserted into our proposed network with less computational complexity. In this article, the attention block is followed by each convolutional operation, whose initial weights are transferred from VGGNet-16. We discuss how attention mechanism facilitates the interpretability of our proposed methods and the distribution of attention values in experiments.

The attention block can be trained with our proposed transfer learning for HSI classification. In this article, we use the cross entropy as the loss function, which is optimized using the Adam method with a backpropagation algorithm. The Adam method is more suitable for calculating posterior probabilities in the case of neural networks, which is defined as follows [64]:

$$\text{Loss} = \frac{1}{C} \sum_{a=1}^{r} \left[-L'_a \log(L_a) - \left(1 - L'_a\right) \log\left(1 - L_a\right) \right] \tag{6}$$

where L'_a is the ground-truth labels, L_a is the model output, and C is the total classes of ground truth, and the loss is averaged across observations for each mini-batch.

IV. Experimental Results

A. Data Description

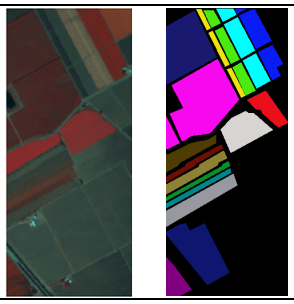
In this article, four widely- used HSIs with different environmental settings were chosen to evaluate the performance of the proposed methods. These data sets are Salinas, Pavia University (Pavia), Kennedy Space Center (KSC), and Indian Pines. The color composite image, the classes with the corresponding number of samples, and the ground truth of four widely used data sets are listed in Fig. 5.

The Salinas data set was collected by the 224-band Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over Salinas Valley, CA, USA. The data set consists of 204 spectral channels after the removal of 20 water absorption bands (108–112, 154–167, and 224). Its spatial resolution is 3.7 m with 512 × 217 pixels. The ground truth contains 16 different classes representing mostly different types of crops.

The Pavia data set was captured by the Reflective Optics Spectrographic Image System (ROSIS) sensor during a flight

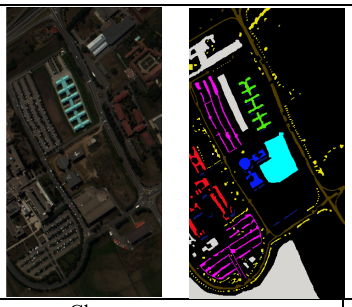
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

Salinas



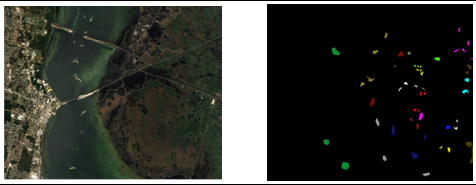
Class			Samples
No.	Color	Name	Numbers
1		Brocoli_green_weeds_1	1977
2		Brocoli_green_weeds_2	3726
3		Fallow	1976
4		Fallow_rough_plow	1394
5		Fallow_smooth	2678
6		Stubble	3959
7		Celery	3579
8		Grapes_untrained	11213
9		Soil_vinyard_develop	6197
10		Corn_senesced_green_weeds	3249
11		Lettuce_romaine_4wk	1058
12		Lettuce_romaine_5wk	1908
13		Lettuce_romaine_6wk	909
14		Lettuce_romaine_7wk	1061
15		Vinyard_untrained	7164
16		Vinyard_vertical_trellis	1737
Total			53785

Pavia



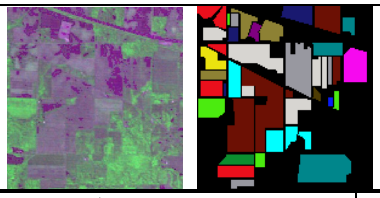
Class			Samples
No.	Color	Name	Numbers
1		Asphalt	6851
2		Meadows	18686
3		Gravel	2207
4		Trees	3436
5		Metal sheets	1378
6		Bare soil	5104
7		Bitumen	1356
8		Bricks	3878
9		Shadow	1026
Total			43922

KSC



Class			Samples
No.	Color	Name	Numbers
1		Scrub	761
2		Willow swamp	243
3		CP hammock	256
4		Slash pine	252
5		Oak/Broadleaf	161
6		Hardwood	229
7		Swamp	105
8		Graminoid marsh	431
9		Spartina marsh	520
10		Cattail marsh	404
11		Salt marsh	419
12		Mud flats	503
13		Water	927
Total			5211

Indian Pines



Class			Samples
No.	Color	Name	Numbers
1		Alfalfa	46
2		Corn-notill	1428
3		Corn-min	830
4		Corn	237
5		Grass-pasture	483
6		Grass-trees	730
7		Grass-pasture-mowed	28
8		Hay-windrowed	478
9		Oats	20
10		Soybean-notill	972
11		Soybean-mintill	2455
12		Soybean-clean	593
13		Wheat	205
14		Woods	1265
15		Buildings-Grass-Trees	386
16		Stone-Steel-Towers	93
Total			10249

Fig. 5. False-color composite, ground reference map, and number of available samples of the four widely used data sets (Salinas, Pavia, KSC, and Indian Pines).

campaign over Pavia, Northern Italy. It is composed of 610 $\times$ 340 pixels with a spatial resolution of 1.3 m/pixel and 115 bands with a spectral coverage ranging from 0.43 to 0.86 μm; 12 noisy spectral bands are removed, leading to 103 channels in the experiments. The ground truth covers nine classes of interest, which represents a typical urban site.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

The KSC data set was captured by the AVIRIS over Florida with an altitude of approximately 20 km. The data set is composed of 512 × 614 pixels with a spatial resolution of 18 m. After removing water absorption and low signal-to-noise ratio bands, the image consists of 176 bands with a spectral coverage ranging from 0.4 to 2.5 μm. The corresponding ground reference data includes 13 classes.

The Indian Pines data set was acquired by the AVIRIS sensor over the Indian Pines test site in Northwestern Indiana. This data set consists of 145 × 145 pixels. This data set contains 224 spectral bands in the wavelength range of 0.2- and 2.4-μm region of the visual and infrared spectrum; 24 spectral bands were removed due to the existence of noise and the remaining 200 bands were used for experiments. The ground truth contains 16 classes.

B. Controlled Random Sampling Strategy

Current spectral–spatial methods generally adopt the random sampling of training samples to evaluate the performance of HSI classification. The random sampling strategy, as its name indicates, randomly samples the training and test samples to ensure that these samples are spatially disjoint. However, some studies have pointed out that the random sampling strategy would overlook the spatial dependence, and therefore, they proposed some alternative sampling strategies.

In [46], a novel sampling strategy was developed, which chooses different local areas in HSI and selects samples continuously from a local area for each class, which greatly reduces the overlap. This method, which is called the controlled random sampling strategy, became a classical strategy in sampling scheme. This sampling strategy mainly consists of the following three steps.

First, a partition is defined by groping connected pixels with the same label. For one class k on the ground truth of HSI, there are several partitions distributed on the ground truth, which means that there are different locations with the same label. All unconnected partitions P are selected in k. All samples n_p are counted in each p.

Second, the seed point d is randomly chosen in p to guarantee training samples distributed globally and randomly. The training samples in each p are defined as

$$n_t = n_p \times s \tag{7}$$

where s is the sampling rate. Here, the region-growing algorithm is applied to extend d to a region, whose size is equal to n_t.

Third, the second step is iterated, all the classes C are finished, and all training samples are selected by combining these samples in each region.

While the controlled random sampling strategy reduces overlap between the training and test samples, to compare with majority of research for proper evaluation, this article mainly adopts the widespread random sampling strategy to validate the proposed methods for HSI classification.

C. Experimental Setup

For all four widely used data sets with the random sampling strategy, the labeled samples are divided into three subsets, including training samples (i.e., 200 samples), validation samples (i.e., 100 samples), and test samples (i.e., the remaining samples). Therefore, the training samples, which are used to learn the weights and biases of each neuron, are truly limited. The validation samples are used to guide the proper network design. All the remaining samples are used to estimate the final classification results of the proposed methods. Besides, for the controlled random sampling strategy, the number of training samples is selected according to 5%, 10%, and 25% sampling rate, and the remaining samples are used as test samples.

The proposed network is shown in Fig. 1. The patches with the size of $33 \times 33 \times B$ are fed to the proposed network as an input. VGGNet-16 is adopted as the pretrained model, which has 13 convolutional layers and mainly uses 3×3 convolutional kernels. Each convolutional layer in VGGNet-16 is followed by padding, a BN layer, and a ReLU operation. One size of padding is used to maintain the size of the original input. The combination of the three layers (i.e., convolutional, BN, and ReLU layers) shapes one convolutional block. The max-pooling layer is used in every two or three convolutional blocks. In our experiments, only seven blocks are used to initialize the weights of the proposed network. The channel mapping uses an FC neural network with one layer. The mapping transformation receives all the bands as an input, while its output is of three features. Therefore, the prepared data with three channels can be used as the input for VGGNet-16.

For HT-CNN, all transfer layers are finely tuned using the labeled training samples. During the training phase, the training epoch is set to 60 for all the data sets used in this article. The learning rate of HT-CNN is set to 0.0005 and 0.008 for the Salinas and Pavia data sets, respectively. The initial learning rate is 0.001 for the KSC and Indian Pines data sets. The learning rate of the KSC and Indian Pines data set is reduced by half after 20 and 40 epochs, respectively. In order to evaluate the performance of the proposed methods quantitatively, overall accuracy (OA), average accuracy (AA), and kappa coefficient (K) are investigated.

D. Classification Results of HT-CNN

1) Comparisons With Different Methods: Here, the performance of the proposed methods is compared with other typical HSI classification approaches. SVM with extended morphological profiles (EMP-SVM) [65] is used. EMP includes the opening and closing operations that can extract the spatial information of HSIs on the first three principal components. In the experiments, we adopt a disk-shaped structuring element with a size increasing from two to eight with the interval of two. Thus, the EMP produces 27 spatial features. Then, the learned spectral–spatial features are fed into RBF-SVM to produce the final classification map.

Fivefold cross validation is used to find the best value of c and γ. The optimal value of c and γ is traced in (10^{-3}, 10^{-2}, 10^{-1}, 10, 10^1, 10^2, and 10^3). The size of the input patches is set to 33×33. CNN is composed of three convolutional layers with different kernel sizes (i.e., 5×5, 4×4, and 3×3), and the corresponding output channel is 32, 48, and 64, respectively. Every convolutional layer is followed by the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

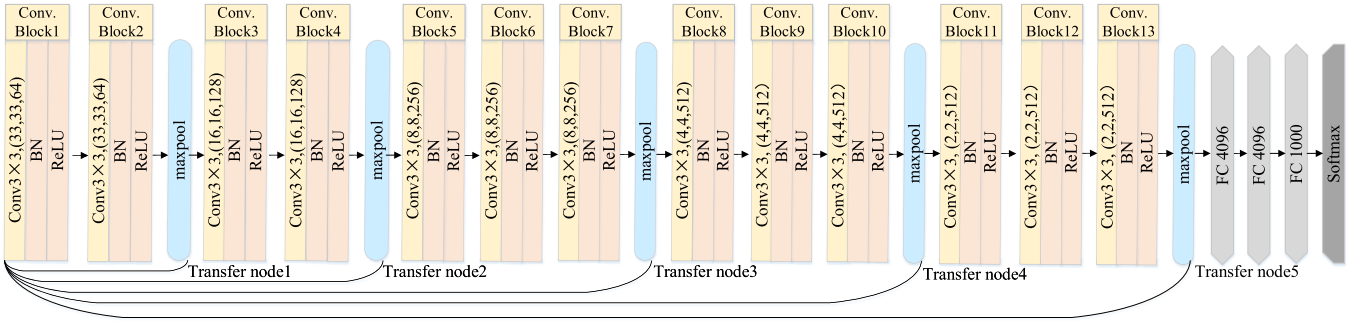


Fig. 6. Source model (i.e., VGGNet-16) for HSI classification.

BN and ReLU operations. Max pooling is only used after the first two ReLU operations. Moreover, we also used the spectral–spatial residual network (SSRN), which was recently proposed for HSI classification [38]. In our experiments, VGG has the architecture of the original VGGNet-16, except for the last three FC layers. Here, VGG is reproduced for the different HSIs. The experiments with RGB channels (RGB-T-CNN) are also presented for comparison. Due to the fact that Salinas, KSC, and Indian Pines data sets were captured by the AVIRIS sensor, three bands (i.e., 36, 19, and 11) are selected, which can approximate red, green, and blue wavelength ranges, respectively. For the Pavia data set, which was captured by ROSIS, band numbers 77, 33, and 12 are chosen.

In order to validate the effectiveness of transfer learning with heterogeneous data sets mapping, we randomly select three bands from the all bands of HSIs, and we call this method as T-CNN for short.

Tables I–IV show all experimental results. All the experiments are run for ten times with different training samples which are randomly selected. We report the classification results in the form of mean ± standard deviation. The proposed method achieves the highest accuracy in terms of OA, AA, and K for all widely used data sets with 200 training samples. For the Salinas data set, HT-CNN increases the OA, AA, and K of T-CNN by 0.2%, 1.08%, and 0.82%, respectively. Moreover, HT-CNN exhibits the best OA over CNN, SSRN, and VGG, with the improvement of 4.97%, 4.64%, and 4.4%, respectively. It can be seen that compared to the CNN-based methods (i.e., CNN and SSRN), EMP-SVM provides competitive classification results. HT-CNN outperforms EMP-SVM by 5.78%, 2.77%, and 6.13% for the Salinas data set in terms of OA, AA, and K, respectively. For the Pavia data set, HT-CNN reaches 93.26%, which is higher than that of T-CNN with improvement of 0.71%, 1.5%, and 0.97% in terms of OA, AA, and K. The performance of the proposed RGB-T-CNN is superior to the CNN. In this context, the RGB-T-CNN increases the OA of the other method by 4.62%, 1.43%, 0.67%, and 2.38% on the Salinas, Pavia, KSC, and Salinas data sets, respectively, with 200 random sampling samples. From the results, one can see that the proposed methods work well even if the number of training samples is very limited for all four widely used HSIs.

2) Analysis of Different Transfer Layers: In this experiment, we investigate the optimal transfer layers of the original

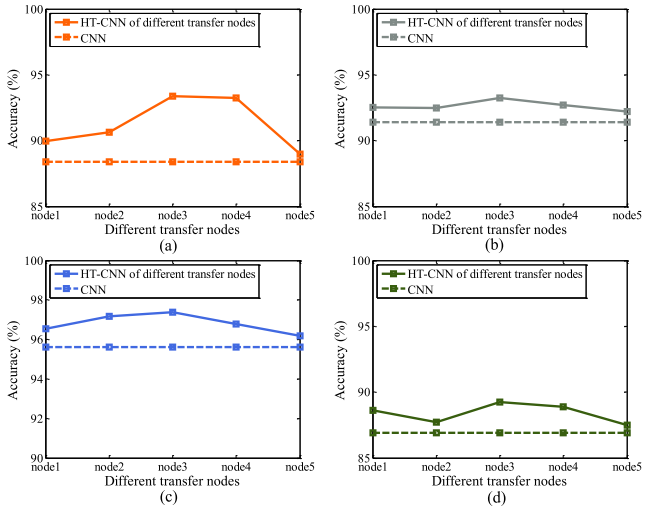


Fig. 7. Classification accuracy of different transfer nodes on the four widely used data sets. (a) Salinas. (b) Pavia. (c) KSC. (d) Indian Pines.

VGGNet-16 (see Fig. 6) for the HSI classification task. It has been proven that the lower level features in hidden layers are universal, but the higher level features are specific for different tasks [66]. The originalVGGNet-16, which is used here, includes 13 convolutional blocks and five transfer nodes (each max-pooling layer is regarded here as a node), which are trained using the ImageNet data set. It is important to select the appropriate transfer nodes for HSI classification. The final classification accuracy is used as the metrics to assess the result of different transfer nodes. The results are compared with CNN and are shown in Fig. 7.

From Fig. 7, which compares CNN and different transfer nodes, it can be seen that different transfer nodes usually have a superior performance on the four widely used HSIs. For example, for the Salinas data set, compared to the CNN, transfer node 3 reaches almost 93%, with the improvement of nearly 5%. All the experimental results prove that the proposed transfer methods are effective. It can be seen that the transfer node 3 always achieves the most accurate results for all four HSIs. Thus, in our experiments, the architecture of transfer node 3 is adopted as the final HT-CNN for HSI classification.

3) Analysis of Fine-Tuning of Transfer Learning: In this section, all layers in HT-CNN are trained simultaneously, which means that the weights in original VGGNet-16 are all fine-tuned. We compare the similarity of the architecture

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

TABLE I
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE SALINAS DATA SET USING 200 TRAINING SAMPLES

Method	EMP-SVM	CNN	SSRN	VGG	RGB-T-CNN	T-CNN	HT-CNN	HT-CNN - Attention
OA(%)	87.59±2.39	88.40±2.13	88.73±2.06	88.97±3.24	93.02±1.42	93.17±1.46	93.37±1.05	**94.70±2.05**
AA(%)	88.65±1.93	92.48±2.79	92.63±1.47	92.83±2.62	91.27±2.40	90.34±2.79	91.42±4.23	**92.37±1.27**
K×100	86.15±2.75	88.27±2.73	87.51±2.26	87.29±4.30	92.06±2.28	92.16±1.79	92.98±1.67	**93.62±2.69**
Brocoli_green_weeds_1	95.28±4.21	81.75±5.36	**100.00±0.00**	85.25±9.45	81.08±16.57	90.25±11.95	92.60±10.01	97.42±5.16
Brocoli_green_weeds_2	98.49±0.28	88.89±7.69	**100.00±0.00**	96.00±5.15	98.54±3.30	99.28±1.45	97.61±2.02	98.85±2.30
Fallow	70.53±22.58	89.35±3.85	72.85±13.98	89.24±8.89	78.54±29.35	71.84±23.63	**90.03±9.44**	75.55±20.30
Fallow_rough_plow	99.80±0.12	80.01±5.86	99.78±2.15	100.00±0.00	93.64±9.39	93.22±6.48	89.73±20.54	**100.00±0.00**
Fallow_smooth	93.34±4.40	88.40±2.57	98.68±0.54	98.46±2.24	**98.78±1.28**	95.69±4.86	96.77±2.18	98.35±1.49
Stubble	99.36±0.22	90.45±6.38	99.85±0.89	**100.00±0.00**	99.37±0.99	96.14±3.22	99.32±0.91	92.50±2.54
Celery	97.48±1.84	97.96±2.79	99.43±0.43	79.68±9.84	99.53±0.72	100.00±0.00	100.00±0.00	**100.00±0.00**
Grapes_untrained	90.09±7.83	69.63±9.62	62.90±14.95	74.50±6.45	94.41±6.17	94.24±3.98	98.87±0.92	**99.22±0.57**
Soil_vinyard_develop	98.89±0.08	89.33±9.79	97.87±3.96	99.67±0.41	99.51±0.85	100.00±0.00	99.85±0.31	**100.00±0.00**
Corn_senesced_green_weeds	90.74±2.23	85.75±9.08	88.08±5.20	96.70±6.00	97.04±5.83	**97.86±1.48**	95.84±7.56	95.28±7.89
Lettuce_romaine_4wk	93.06±2.02	88.92±9.72	86.54±7.17	83.82±6.70	97.65±7.06	93.15±10.85	90.48±15.91	**98.75±2.50**
Lettuce_romaine_5wk	**99.98±0.04**	82.07±9.37	99.41±5.75	98.91±1.34	98.41±3.39	98.78±1.49	97.34±2.64	99.58±0.83
Lettuce_romaine_6wk	77.91±39.03	82.65±8.30	84.74±7.69	93.57±9.69	93.79±9.62	76.05±33.94	**96.00±8.00**	66.05±35.15
Lettuce_romaine_7wk	98.34±0.99	85.41±9.25	99.33±1.18	79.13±9.60	82.84±22.16	95.83±5.27	**99.13±1.74**	74.12±37.74
Vinyard_untrained	35.73±30.40	76.80±18.48	**93.40±12.17**	87.21±11.03	75.42±16.12	81.07±16.68	68.84±5.81	84.25±7.37
Vinyard_vertical_trellis	79.35±6.22	56.06±12.02	**98.71±2.58**	75.14±17.03	71.75±29.73	62.03±23.75	50.30±33.68	57.26±36.04
Training Time (sec.)	2.27	28.99	141.57	63.62	9.00	7.32	18.08	22.44
Test Time (sec.)	1.03	82.52	66.69	88.43	5.61	5.94	84.91	88.48

TABLE II
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE PAVIA DATA SET USING 200 TRAINING SAMPLES

Method	EMP-SVM	CNN	SSRN	VGG	RGB-T-CNN	T-CNN	HT-CNN	HT-CNN - Attention
OA(%)	89.43±1.20	91.41±1.44	91.59±3.57	92.21±1.40	92.84±1.83	92.55± 2.06	93.26±1.18	**94.25±1.08**
AA(%)	80.37±3.60	81.03±4.99	87.56±3.57	87.23±2.52	84.71±3.60	84.84±2.43	86.34±2.65	**87.86±2.45**
K×100	85.81±1.64	89.12±1.83	88.963±1.56	90.19±1.76	90.86±2.44	90.54±2.64	91.51±1.49	**92.36±2.41**
Asphalt	91.84±1.75	92.36±5.62	**99.62±2.57**	88.77±7.01	91.97±3.74	92.59±4.48	92.51±5.41	92.42±3.97
Meadows	98.02±0.68	98.85±1.04	98.85±4.91	98.24±1.14	**99.83±0.14**	99.66±0.19	99.15±0.58	99.44±0.50
Gravel	66.63±14.10	42.94±22.88	**90.99±25.41**	62.73±6.89	75.49±22.43	70.42±12.74	73.48±16.79	76.38±11.62
Trees	92.10±5.14	93.52±3.53	**94.13±8.55**	86.97±8.19	74.38±10.80	72.63±11.76	81.37±6.70	84.31±8.96
Metal sheets	86.26±21.92	99.33±0.72	**99.63±2.74**	96.18±4.32	91.77±20.49	95.84±3.54	92.89±11.38	97.93±2.93
Bare soil	78.66±7.12	98.62±1.54	77.72±8.91	94.99±2.81	98.81±1.67	98.62±2.29	**98.81±2.01**	97.66±4.67
Bitumen	74.43±10.12	45.56±27.30	59.63±15.34	82.13±14.19	78.39±17.68	82.02±12.91	**84.92±9.39**	89.48±8.07
Bricks	90.92±3.70	90.60±5.68	67.52±23.28	93.01±2.69	**97.33±1.81**	95.35±4.84	95.31±3.43	96.35±2.00
Shadow	74.11±22.21	67.52±33.51	**99.90±2.89**	82.05±4.52	54.39±27.07	56.41±27.04	58.63±22.06	56.76±32.92
Training Time (sec.)	3.52	18.52	76.82	26.56	12.62	9.73	17.13	19.29
Test Time (sec.)	0.14	28.77	20.58	30.61	6.74	4.71	30.98	34.02

with and without fine-tuning (the initialization of mapping transformation is retained the same) of different transfer nodes. Here, CKA with RBF 0.5 [see (5)] is used to evaluate different features, which is recently proposed to measure the relationship between the networks trained with different experimental settings.

The features after each convolutional block in HT-CNN are computed by CKA. For each convolutional block of HT-CNN with and without fine-tuning, the similarity of the corresponding convolutional layer is computed by RBF CKA, which reflects the fraction of the median Euclidean distance. In Fig. 8, we list the results of CKA for four HSIs, and the bar in each figure represents the similarity of learned features in HT-CNN with and without fine-tuning.

From Fig. 8, it can be seen that the values of CKA in the first several convolutional blocks are higher compared

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

TABLE III
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE KSC DATA SET USING 200 TRAINING SAMPLES

Method	EMP-SVM	CNN	SSRN	VGG	RGB-T-CNN	T-CNN	HT-CNN	HT-CNN - Attention
OA(%)	93.26±1.65	95.62±0.95	96.55±0.42	96.19±1.52	96.29±1.60	96.14±1.69	97.39±0.27	**98.31±0.80**
AA(%)	89.62±3.04	92.82±1.41	94.43±0.45	94.47±2.33	94.40±2.83	94.03±3.40	95.98±0.55	**97.47±1.00**
K×100	92.49±1.84	95.12±1.06	96.17±0.46	95.75±1.70	96.10±1.65	95.69±1.89	97.10±0.30	**98.12±0.89**
Scrub	96.17±3.98	98.05±3.49	98.40±1.70	95.08±5.96	97.43±7.08	100.00±0.00	100.00±0.00	**100.00±0.00**
Willow swamp	87.81±11.27	89.25±6.26	88.60±11.66	89.18±9.65	70.41±16.87	74.09±25.86	87.76±10.19	**90.96±10.38**
CP hammock	82.76±11.31	92.62±13.37	94.85±2.22	93.89±8.38	91.40±10.48	89.15±6.83	90.26±5.63	**97.83±3.36**
Slash pine	74.98±13.85	83.34±12.82	95.99±3.20	78.20±3.20	91.94±9.56	**97.04±3.27**	92.70±7.89	86.77±9.41
Oak/Broadleaf	70.79±7.68	66.95±13.73	78.95±6.99	90.57±16.84	94.88±6.72	92.24±9.24	84.79±9.38	**94.93±7.32**
Hardwood	83.93±9.35	90.94±8.36	88.93±10.23	85.06±15.49	**99.77±0.68**	96.41±6.28	95.71±7.26	97.73±3.69
Swamp	87.67±26.15	92.45±9.54	92.79±11.85	100.00±0.00	86.93±23.76	84.03±19.87	**100.00±0.00**	99.80±0.40
Graminoid marsh	93.14±7.53	93.88±5.34	93.51±5.82	97.04±4.65	96.21±6.37	90.40±5.67	96.91±6.04	**99.03±1.20**
Spartina marsh	94.74±6.42	99.67±0.56	98.40±3.20	99.05±1.36	99.84±0.47	100.00±0.00	100.00±0.00	**100.00±0.00**
Cattail marsh	94.95±4.70	100.00±0.00	99.68±0.65	100.00±0.00	99.34±1.14	100.00±0.00	100.00±0.00	**100.00±0.00**
Salt marsh	99.97±0.08	100.00±0.00	97.73±4.53	100.00±0.00	99.58±0.69	99.95±0.10	100.00±0.00	**100.00±0.00**
Mud flats	98.28±1.81	99.49±0.64	97.79±0.30	100.00±0.00	99.51±1.18	99.11±1.19	99.66±0.44	**100.00±0.00**
Water	99.93±0.06	100.00±0.00	100±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
Training Time (sec.)	4.75	17.19	116.43	30.71	12.67	9.76	15.35	24.84
Test Time (sec.)	0.07	6.38	1.88	7.09	0.85	0.49	2.77	3.08

TABLE IV
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE INDIAN PINES DATA SET USING 200 TRAINING SAMPLES

Method	EMP-SVM	CNN	SSRN	VGG	RGB-T-CNN	T-CNN	HT-CNN	HT-CNN - Attention
OA(%)	81.53±2.39	86.81±1.28	83.21±1.25	87.49±1.10	89.19±1.26	87.83±0.78	89.24±1.47	**90.86±0.98**
AA(%)	72.43±6.35	68.30±4.82	62.88±7.12	65.69±2.46	70.12±6.87	65.32±2.77	70.47±4.50	**74.68±8.75**
K×100	78.87±2.78	84.20±1.57	80.86±1.39	84.94±1.35	86.92±1.93	85.35±1.00	87.06±1.74	**89.05±1.19**
Alfalfa	50.27±41.37	38.70±32.39	25.00±35.08	36.69±36.63	**53.64±43.99**	37.78±36.95	40.26±37.94	39.06±47.84
Corn-notill	72.43±7.51	84.64±6.07	87.16±12.44	88.58±5.30	92.13±4.43	93.03±3.61	89.12±5.43	**93.81±4.75**
Corn-min	80.77±5.42	63.15±22.10	**95.61±20.25**	41.56±19.45	51.61±19.00	45.46±17.23	59.43±12.18	60.52±20.77
Corn	74.44±19.28	55.37±45.79	20.60±17.82	50.74±37.13	**81.46±18.55**	34.79±33.53	46.00±39.09	41.96±34.33
Grass-pasture	77.04±8.10	56.41±10.40	**84.93±21.91**	51.77±11.14	84.48±10.98	82.07±10.06	64.32±13.13	68.53±16.46
Grass-trees	95.27±3.54	96.90±3.94	**99.71±6.22**	92.58±5.59	95.96±3.39	94.02±4.19	90.38±4.84	95.86±1.11
Grass-pasture-mowed	49.63±49.64	42.54±38.16	28.70±48.04	15.56±20.74	25.17±35.12	38.52±32.84	28.89±35.40	**74.07±38.77**
Hay-windrowed	97.43±2.90	85.84±27.28	**97.65±2.15**	86.20±9.72	73.30±20.69	70.01±33.41	93.55±4.39	86.38±25.18
Oats	28.95±39.68	12.75±18.33	9.28±43.22	8.89±17.78	22.50±36.14	13.33±26.67	43.16±34.50	**48.42±41.23**
Soybean-notill	74.92±7.72	90.33±6.48	80.77±8.61	91.50±2.79	93.45±3.85	89.20±6.52	92.77±5.03	**93.56±4.62**
Soybean-mintill	85.30±4.85	94.21±3.32	83.09±6.57	96.28±1.34	93.43±3.66	94.45±2.11	**97.90±2.00**	95.69±1.16
Soybean-clean	60.49±10.01	58.07±12.47	70.77±21.72	62.57±25.65	80.58±22.81	77.95±21.29	78.10±16.83	**87.93±7.87**
Wheat	75.54±37.88	85.99±14.62	98.48±12.96	88.29±10.88	77.62±25.82	92.61±8.42	92.62±10.15	**96.49±3.88**
Woods	96.56±3.24	96.63±2.53	96.99±3.86	97.63±1.98	**98.23±2.88**	93.05±4.13	94.17±4.40	95.07±1.16
Buildings-Grass-Trees	73.77±9.55	67.38±28.38	38.76±18.17	**81.58±5.76**	55.24±35.39	36.49±28.41	56.97±36.71	66.50±31.76
Stone-Steel-Towers	**66.09±43.03**	63.96±33.59	53.85±44.51	60.62±19.22	43.14±42.28	52.31±30.74	59.83±34.69	51.08±32.61
Training Time (sec.)	1.40	23.50	132.51	32.28	12.66	9.67	23.12	26.92
Test Time (sec.)	0.22	9.23	3.68	10.32	1.28	0.77	4.82	5.33

with the last several convolutional blocks. With increased convolutional blocks, the similarity between features with and without fine-tuning gradually decreases for the four HSIs. For example, for the Pavia data set, the features after the first convolutional block are most relevant, which achieve 0.99, but in the upper convolutional block, the similarity is only 0.5.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

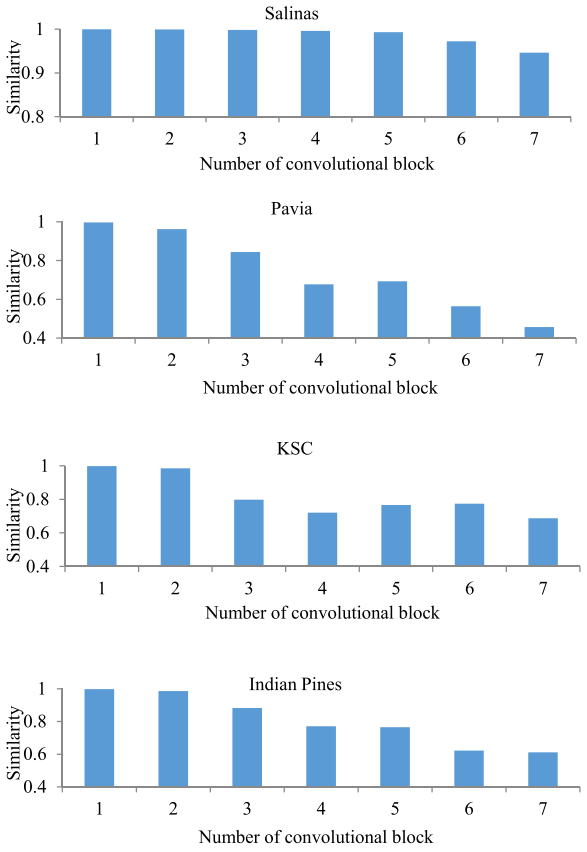


Fig. 8. CKA of different convolutional blocks for transfer node 3 on the four widely used data sets (Salinas, Pavia, KSC, and Indian Pines).

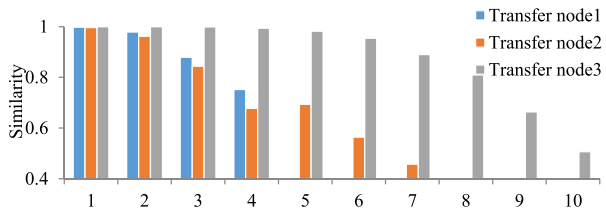


Fig. 9. CKA of different transfer nodes on the Pavia data set.

The reason is that the features of original VGGNet-16 in the lower convolutional blocks are more general, and the similarity of HT-CNN with and without fine-tuning is high. However, as the number of the convolutional blocks increases, transferred features are tend to the specific task, which cause that the similarity of HT-CNN with and without fine-tuning is low.

We also compare the similarity of different transfer nodes with and without fine-tuning. CKA is used to compare the similarity of architectures of different transfer nodes with and without fine-tuning. Here, we chose transfer nodes 1–3 for comparisons. The Pavia data set is used, and the results are shown in Fig. 9. From Fig. 9, it can be seen that the similarity of different transfer nodes with and without fine-tuning is similar, which always has a negative trend as the number of convolutional blocks increases.

Furthermore, the well-designed architecture has experimented on four widely used data sets with another sampling strategy (i.e., the controlled random sampling strategy), which is reported in Section IV-F.

E. Classification Results for HT-CNN-Attention

In this section, we report the effectiveness of the proposed HT-CNN with attention mechanism, which is named HT-CNN-Attention.

Feature maps in each convolutional layer with transfer learning learn different features, which lead to various importance for the transferred feature maps. Therefore, to narrow the gap between the two data sets, attention mechanism is used to measure these features maps for better classification results. Each feature map has a value to reflect its importance, which we call attention value.

1) Experiential Design: We conduct experiments for HT-CNN and HT-CNN-Attention with the same parameter settings (i.e., learning rate and batch size). Then, each attention block is inserted after each convolution block of HT-CNN. The attention block includes an average pooling operation to capture the global information embedding and two small FC layers for adaptive recalibration. To balance a tradeoff between performance and computational cost in HT-CNN-Attention, the reduction ratio (r) is set to 16. To partially address the overfitting problem during the training process, dropout is used before the FC layer. To evaluate the performance of HT-CNN-Attention effectively, the hyperparameters (i.e., learning rate, batch size, and learning rate) are set to be the same as those in HT-CNN. Tables I–IV show the classification results with 200 training samples. The results of different training samples (i.e., 300 and 400) for four HSIs are also reported in Table V. The convergence curves of the two architectures with 200 training samples and 100 validation samples for the four HSIs are shown in Fig. 10.

2) Results of HT-CNN-Attention: From Tables I–V, it can be seen that HT-CNN-Attention leads to performance improvements. Compared with HT-CNN, the OA of HT-CNN-Attention increases by 1.33%, 0.99%, 0.92%, and 1.62% with 200 training samples on the Salinas, Pavia, KSC, and Indian Pines, respectively. Furthermore, the HT-CNN-Attention with 300 and 400 training samples also demonstrates the effectiveness of our methods. The best results are obtained using HT-CNN-Attention with 400 training samples, which reached 99.00%, 97.78%, 99.78%, and 96.99% for the Salinas, Pavia, KSC, and Indian Pines data sets, respectively. Compared with 300 training samples, the difference in OA between HT-CNN and HT-CNN-Attention is smaller with 400 training samples. For example, for the Salinas data set, when the number of training samples is 400, the OA of HT-CNN-Attention improves the OA of HT-CNN by 0.27%, which demonstrates that the HT-CNN-Attention is able to work more effectively with limited training samples.

The training curves for HT-CNN and their respective HT-CNN-Attention are described in Fig. 10. It can be observed that HT-CNN-Attention yields a steady improvement throughout the training process for all four HSIs.

3) Visualization and Analysis Attention Values of HT-CNN-Attention: In this experiment, we study the distribution of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

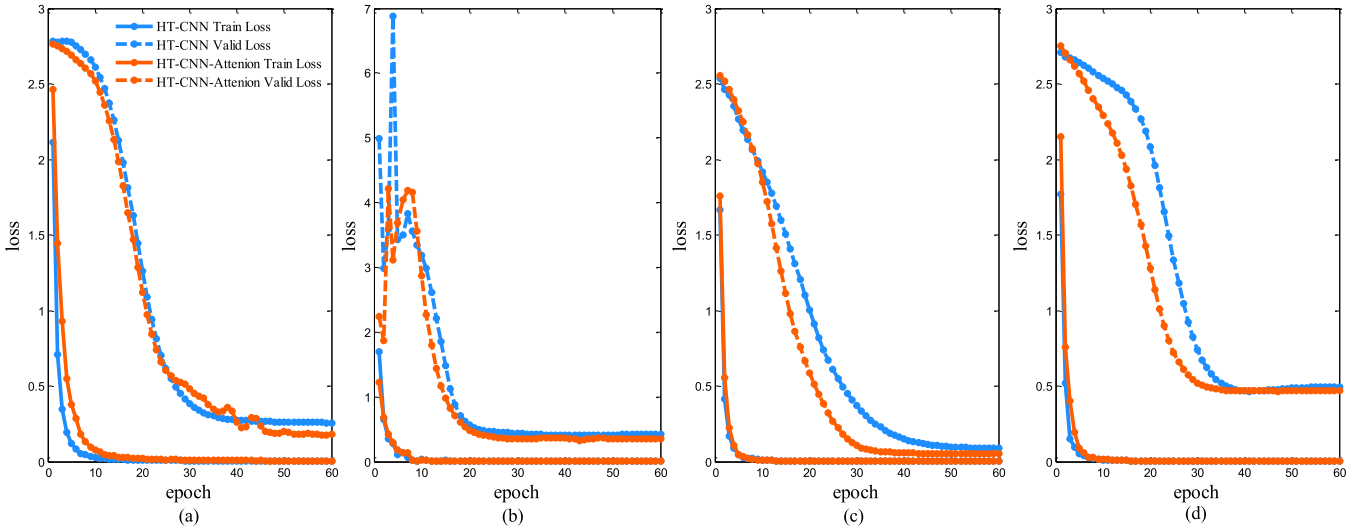


Fig. 10. Convergence curve comparisons for HT-CNN and HT-CNN-Attention on the four widely used data sets. (a) Salinas. (b) Pavia. (c) KSC. (d) Indian Pines.

TABLE V

CLASSIFICATION RESULTS (VALUES) ON THE FOUR WIDELY USED DATA SETS (SALINAS, PAVIA, KSC, AND INDIAN PINES) WITH 300 OR 400 TRAINING SAMPLES

Datasets / Class		Salinas				Pavia				KSC				Indian Pines			
		300 Training Samples		400 Training Samples		300 Training Samples		400 Training Samples		300 Training Samples		400 Training Samples		300 Training Samples		400 Training Samples	
No.	Color	HT-CNN	HT-CNN-Attention	HT-CNN	HT-CNN-Attention	HT-CNN	HT-CNN-Attention	HT-CNN	HT-CNN-Attention	HT-CNN	HT-CNN-Attention	HT-CNN	HT-CNN-Attention	HT-CNN	HT-CNN-Attention	HT-CNN	HT-CNN-Attention
1		90.97	**100**	100	**100**	**96.43**	96.02	**98.35**	96.79	99.91	**100**	100	**100**	41.08	**52.08**	66.31	**93.02**
2		98.72	**100**	100	**100**	98.70	**99.89**	99.11	**99.83**	97.40	**99.07**	**99.11**	98.06	**96.08**	92.94	**97.30**	96.50
3		**93.47**	87.15	**100**	99.07	79.28	**87.77**	94.55	**95.57**	**97.45**	97.21	**99.91**	98.63	70.11	**79.79**	**87.46**	87.12
4		97.57	**100**	99.23	**100**	**85.69**	83.33	88.25	**93.79**	91.72	**96.20**	98.04	**100**	83.02	**85.42**	84.75	**85.92**
5		**98.98**	98.91	98.88	**99.07**	97.13	**97.55**	**99.57**	99.26	97.75	**100**	96.04	**99.59**	84.26	**88.56**	95.93	**97.78**
6		100	**100**	100	**100**	96.65	**99.72**	**99.81**	99.64	99.25	**99.44**	100	**100**	97.91	**97.99**	98.48	**99.26**
7		99.06	**99.70**	100	**100**	92.43	**92.99**	**95.49**	93.15	100	**100**	100	**100**	71.03	**73.08**	71.68	**89.52**
8		98.08	**99.35**	**99.57**	99.56	96.35	**98.32**	97.35	**97.86**	99.487	**100**	**100**	99.71	77.01	**98.84**	**99.89**	99.57
9		100	**100**	100	**100**	72.29	**88.04**	83.29	**85.94**	99.88	**100**	100	**100**	24.80	**61.11**	**53.86**	25.03
10		**99.73**	98.28	**100**	99.62					100	**100**	100	**100**	93.49	**94.05**	**96.76**	95.52
11		**99.13**	78.82	100	**100**					100	**100**	100	**100**	97.26	**98.86**	**98.39**	98.17
12		97.71	**98.43**	99.29	**99.49**					99.45	**100**	100	**100**	**90.02**	81.97	85.58	**95.26**
13		**100**	99.17	95.49	**99.00**					100	**100**	100	**100**	**95.67**	91.39	**97.22**	93.69
14		86.15	**98.67**	100	**100**									**97.03**	97.01	98.63	**99.72**
15		93.51	**95.71**	93.27	**97.80**									79.93	**91.71**	85.79	**89.98**
16		**86.03**	85.37	**65.50**	53.59									80.98	**84.37**	**80.31**	77.35
OA(%)		97.38	**97.78**	98.73	**99.00**	94.91	**96.58**	97.25	**97.78**	99.03	**99.56**	99.70	**99.78**	93.40	**94.33**	96.24	**96.99**
AA(%)		96.19	**96.22**	96.95	**96.70**	90.55	**93.74**	95.09	**95.76**	98.64	**99.38**	99.47	**99.69**	79.98	**85.57**	87.40	**89.15**
K× 100		97.17	**97.87**	98.41	**98.90**	93.58	**95.64**	96.54	**97.20**	98.92	**99.51**	99.66	**99.76**	92.09	**93.21**	95.52	**96.40**

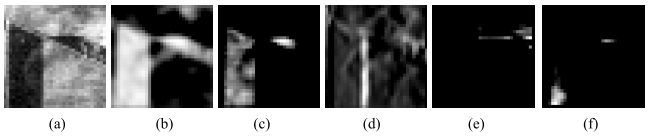


Fig. 11. Feature maps in different channels after the first convolutional layer. (a) Test sample. (b) Tenth feature map. (c) 20th feature map. (d) 30th feature map. (e) 40th feature map. (f) 50th feature map.

attention values in each feature map to provide clearer pictures of HT-CNN-Attention. Different feature maps capture different features in each convolutional layer (as shown in Fig. 11, we select several feature maps in the first convolutional block as an example), which cause different importance of these feature maps. Thus, HT-CNN-Attention is proposed to weigh different feature maps, which consists of three steps. First, the global average pooling operation is used to obtain global information of all feature maps. Then, global information is recalibrated to form different attention values through two nonlinearly operations and a ReLU operation. Finally, the attention values are normalized within [0, 1] by the sigmoid function, which are, then, multiplied by each feature map. Here, we investigate different attention values of each feature map and plot the curves of each attention block.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

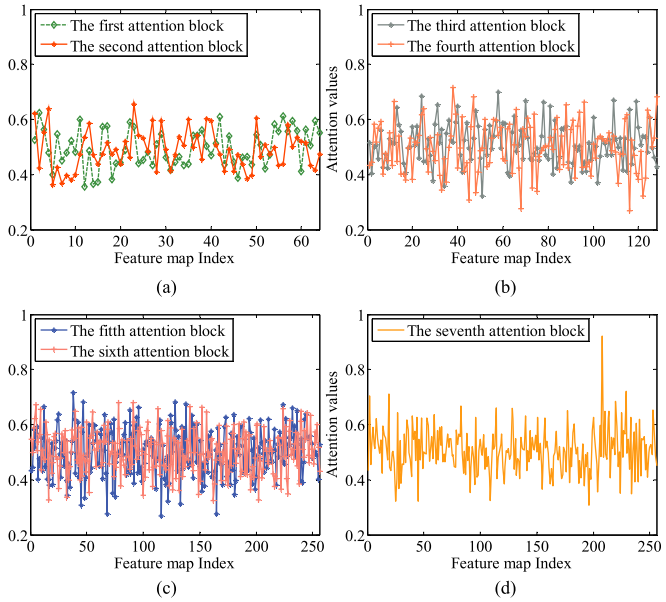


Fig. 12. Distribution of attention values in each attention block. (a) First and second attention blocks. (b) Third and fourth attention blocks. (c) Fifth and sixth attention blocks. (d) Seventh attention block.

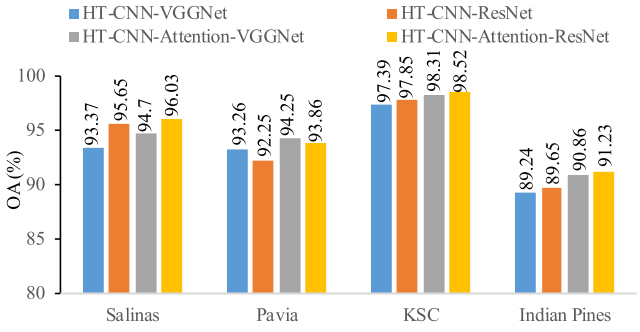


Fig. 13. Proposed methods with two pretrained models (VGGNet-16 and ResNet34) on the four widely used data sets (Salinas, Pavia, KSC, and Indian Pines) with 200 training samples.

In the experiments, the attention block is inserted after each convolutional block in HT-CNN. We choose the Indian Pines data set as an example. The test samples of each class in this example are 29 to draw the distributions of attention values, which are shown in Fig. 12. From Fig. 12, it can be seen that the attention values are different in each attention block, which represents different importance of each feature map. In the first few attention blocks, the distribution of attention values is quite different, such as Fig. 12(a) and (b), which illustrates that the attention block is more important in the first few convolutional blocks. These feature maps in shallow layers extract general features, which need attention blocks to select important features for high-level feature extraction.

4) Comparison of the Proposed Network With Other Pretrained Models: The heterogeneous transfer learning is also tested by considering the ResNet-34 model [43]. The experimental results show that the proposed classification framework is also effective. The experimental results with the pretrained network (ResNet-34) for the proposed methods (i.e., HT-CNN and HT-CNN-Attention) are shown in Fig. 13.

Compared with the VGGNet-16 pre-trained model, it can be seen that HT-CNN and HT-CNN-Attention, which adopt the ResNet-34 pre-trained model, obtain better classification results for the KSC and Indian Pines data sets, respectively, and has worse performance on the Pavia data set. All the proposed methods lead to performance improvement compared to the other studied methods (i.e., CNN, VGG, and SSRN). This demonstrates that in addition to the transfer learning with the VGGNet, transfer with other pretrained model (i.e., ResNet) can also enhance the classification performance.

F. Transfer Learning With Controlled Random Sampling Strategy

In this section, we conduct experiments on the four data sets using another sampling strategy, which is called the controlled random sampling strategy [46] to validate the effectiveness of our proposed methods.

1) Controlled Sampling Strategy: This method becomes a classical strategy in sampling scheme, which borrowed the idea of region growing to create region-shape training samples. The training samples selected by this strategy concentrated locally and dispersed globally, which makes the training samples clustered and spatially distributed [46].

The controlled random sampling strategy alleviates the problem of the traditional random sampling strategy and reduces the influence of overlap between the training and test data, which eventually leads to a fair way for spectral–spatial classification of HSI.

2) Parameter Settings and Results: In order to provide a fair comparison, the controlled random sampling strategy and the number of training samples (e.g., 5%, 10%, and 25% sampling rate) are set in a similar way to the work of Liang *et al.* [46]. To further study the effectiveness of different feature representation methods with this sampling strategy, the methods, including EMP-SVM and CNN, are selected for comparison and the experimental settings are designed according to Section IV-D to maintain the consistency. To mitigate the overfitting problem, the CNN uses the L2 regularization method, and HT-CNN and HT-CNN-Attention use global average pooling. The classification results are shown in Tables VI and VII, and Fig. 14. These results show that the proposed classification methods lead to improved classification accuracies.

The accuracy for each class is important to evaluate the performance of the proposed method. Tables VI and VII report the class-specific accuracies obtained by different methods with the 25% sampling rate. For the KSC data set, as can be seen, all the proposed methods (i.e., RGB-T-CNN, T-CNN, HT-CNN, and HT-CNN-Attention) show higher OA than 89% for all classes, but the lowest accuracy of the CNN is 84.42% belonging to Hardwood. In addition, HT-CNN improves the OA of the EMP-SVM by 1.8%, 3.58%, 12.56%, and 2.01% for Salinas, Pavia, KSC, and Indian Pines data sets, respectively. Besides, HT-CNN-Attention improves the OA of the CNN by 2.08%, 2.79%, 1.51%, and 2.11% for the four widely used data sets, respectively. All the results can verify the effectiveness of the proposed methods with transfer learning. Fig. 14 shows the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

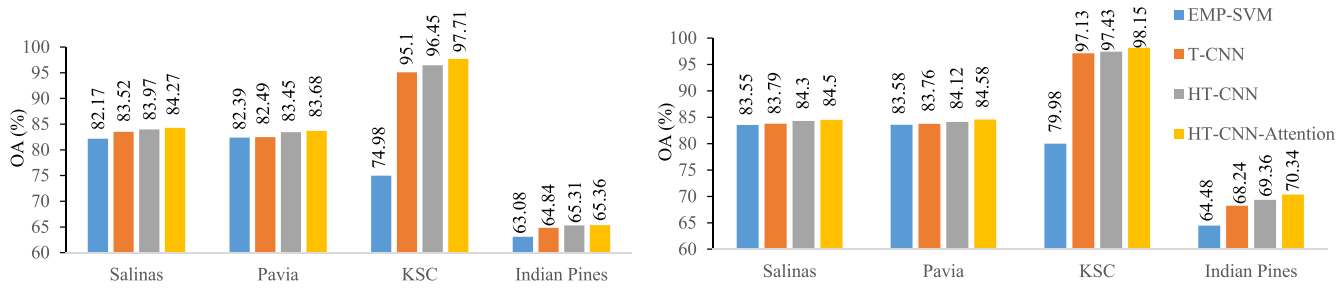


Fig. 14. Classification results of different methods on the four widely used data sets (Salinas, Pavia, KSC, and Indian Pines) with 5% and 10% sampling rate.

TABLE VI

CLASSIFICATION RESULTS (VALUES) ON THE SALINAS AND PAVIA DATA SETS WITH 25% SAMPLING RATE

Color	Salinas						Pavia					
	EMP-SVM	CNN	RGB-T-CNN	T-CNN	HT-CNN	HT-CNN-Attention	EMP-SVM	CNN	RGB-T-CNN	T-CNN	HT-CNN	HT-CNN-Attention
	93.31±9.73	**100±0.00**	100.00±0.00	99.72±0.32	99.80±0.45	98.62±1.58	**91.82±3.98**	79.94±17.37	68.93±10.29	73.84±14.61	77.76±11.89	71.67±15.26
	96.54±5.14	76.40±5.22	90.18±6.71	92.00±5.02	93.60±9.65	**97.64±2.44**	89.97±3.58	94.33±4.53	98.19±0.80	98.88±0.75	**99.04±0.52**	97.53±1.79
	90.02±11.90	66.10±8.09	76.02±16.27	81.83±11.55	**92.70±4.71**	81.56±10.21	83.16±5.60	91.63±6.36	99.57±0.45	**99.62±0.52**	99.60±0.48	99.45±0.81
	98.25±0.86	**100±0.00**	96.60±6.05	99.09±1.37	98.80±3.76	97.57±3.81	96.74±6.82	99.60±0.33	99.04±1.10	99.37±0.51	99.48±0.48	**99.51±0.24**
	94.09±11.84	**99.70±1.78**	98.05±1.75	98.05±1.84	95.90±1.37	98.90±0.86	99.90±1.55	**99.90±0.06**	98.43±1.08	98.16±1.18	99.30±0.58	99.42± 0.52
	97.80±4.92	100±0.00	96.85±3.94	98.04±3.63	**100±0.00**	97.51±2.90	66.90±16.92	76.17±9.12	89.98±5.08	75.42±9.89	73.65±14.48	**90.90±16.42**
	98.18±2.06	95.50±1.81	98.16±2.10	97.72±2.76	98.30±2.10	**98.55±1.57**	**99.86±13.01**	64.41±13.56	85.78±7.32	82.93±7.40	79.51±10.92	88.28±5.42
	84.48±8.90	62.00±13.95	69.87±15.17	70.21±14.62	76.50±13.23	78.98±12.76	88.03±3.89	96.58±3.67	96.18±1.19	96.90±0.99	**97.07±1.01**	94.75±2.71
	98.66±1.70	95.90±3.59	96.81±4.77	96.97±3.46	97.60±0.88	**99.89±0.19**	98.78±0.99	98.81±1.14	98.51±1.28	98.76±0.63	**99.32±0.30**	96.29±1.89
	82.92±12.08	**100±0.00**	95.46±2.97	96.44±2.30	93.30±1.71	92.20±5.37						
	86.24±9..01	**100±0.00**	99.85±0.23	98.43±2.35	100±0.00	99.65±0.64						
	95.44±8.08	**100±0.00**	98.42±1.37	96.30±6.79	100±0.00	99.32±0.60						
	95.03±10.38	**97.80±1.02**	95.36±6.31	84.44±10.61	69.20±9.48	94.52±5.34						
	98.54±1.50	92.00±2.90	74.99±11.64	71.46±25.36	43.00±17.86	69.70±16.73						
	61.60±20.26	**98.20±8.75**	82.00±10.49	81.46±6.35	76.50±7.71	69.05±18.20						
	92.33±9.85	93.40±18.93	93.35±6.31	100.00±0.00	**100±0.00**	97.34±3.47						
OA(%)	86.70±4.70	87.04±3.99	87.55 ± 3.60	87.84±2.94	88.50±3.16	**89.12 ±3.52**	87.58±1.90	89.25±3.95	91.80 ± 1.50	90.81±2.49	91.16±2.99	**92.04 ± 2.67**
AA(%)	90.76±1.00	92.32±2.41	91.37 ± 1.20	91.39±2.18	89.71±1.80	**91.94 ±2.62**	88.12±1.84	89.04±3.36	92.74 ± 1.14	91.54±2.00	91.64±2.59	**93.09 ± 1.90**
K× 100	85.21±5.16	85.65±4.41	86.17 ± 3.97	86.50±3.24	87.24±3.44	**87.84 ±3.93**	84.47±2.22	86.49±4.93	89.74 ± 1.88	88.42±3.13	88.84±3.79	**90.03 ± 3.40**

TABLE VII

CLASSIFICATION RESULTS (VALUES) ON THE KSC AND INDIAN PINES DATA SETS WITH 25%SAMPLING RATE

Color	KSC						Indian Pines					
	EMP-SVM	CNN	RGB-T-CNN	T-CNN	HT-CNN	HT-CNN-Attention	EMP-SVM	CNN	RGB-T-CNN	T-CNN	HT-CNN	HT-CNN-Attention
	73.92±8.40	100.00±0.00	99.92±0.15	98.80±0.47	**100.00±0.00**	99.65±0.47	68.21±29.52	81.18±15.41	85.88±12.81	87.65±7.98	**93.53±5.06**	77.65±6.60
	76.79±4.35	96.24±3.88	99.34±0.88	98.90±0.54	98.48 ±1.35	**99.66±0.68**	47.70±8.90	66.38±11.58	68.52±9.78	71.03±11.08	68.75±8.96	**75.62±5.17**
	53.59±1.91	99.79±0.42	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**	56.96±13.45	73.67±10.69	**88.09±5.36**	83.31±10.66	80.85±13.97	75.96±6.19
	79.41±5.51	95.72±3.24	100.00±0.00	**100.00±0.00**	99.89±0.21	99.89±0.21	57.09±20.95	67.83±8.85	88.70±4.64	81.30±9.49	**89.13±3.89**	81.74±9.28
	96.26±4.48	88.26±1.84	98.18±2.24	98.30±2.47	**99.83±0.33**	99.67±0.66	74.27±10.67	74.33±11.17	95.81±2.05	**96.33±3.17**	95.11±2.35	96.09±3.28
	76.14±3.20	84.42±13.54	99.65±0.70	**100.00±0.00**	97.33±4.28	99.07±1.59	**96.75±2.12**	81.14±8.11	84.06±4.15	82.86±1.56	85.58±1.86	85.36±3.80
	99.09±2.03	97.97±1.72	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**	86.01±17.22	96.19±4.67	95.24±6.02	93.33±9.33	**99.05±1.90**	89.52±12.20
	79.18±6.64	96.63±4.03	95.46±7.62	89.00±4.18	**96.49±5.71**	95.05±5.29	**97.57±1.39**	75.54±17.94	58.77±18.76	61.23±18.93	73.08±18.13	59.69±22.91
	94.65±6.29	99.85±0.31	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**	**85.49±22.91**	74.67±25.44	80.00±25.30	66.67±22.31	80.00±25.30	72.00±20.83
	89.98±5.88	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**	**78.66±8.16**	49.19±9.90	55.64±19.45	54.21±16.11	55.67±19.61	51.64±23.77
	100±0.00	**100.00±0.00**	98.85±1.10	96.20±1.59	98.09± 2.18	99.87±0.16	**75.77±8.03**	74.16±8.63	68.12±4.95	66.73±4.95	69.12±5.25	74.29±6.06
	97.99±2.48	99.26±1.49	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**	59.95±12.51	66.72±10.51	86.97±5.78	**89.26±4.68**	87.38±4.20	82.46±8.97
	97.05±1.19	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**	**91.63±12.62**	89.48±3.61	82.00±10.99	83.77±4.27	84.96±8.72	85.16±9.69
							95.89±3.94	88.77±9.93	85.58±7.43	89.91±3.71	84.73±7.75	79.80±9.68
							94.21±8.06	79.40±13.23	79.40±18.24	74.63±17.91	78.51±20.08	71.34±22.23
							99.66±0.76	76.21±11.82	51.03±13.25	58.28±15.67	65.86±12.40	61.03±8.56
OA(%)	86.73±1.18	97.96±1.13	99.40 ± 0.78	98.56±0.44	99.29±0.51	**99.47 ± 0.43**	71.70±5.62	72.40 ± 5.05	72.46 ± 3.94	73.28±4.07	73.71±4.85	**74.51 ±5.64**
AA(%)	81.16±1.65	96.78±1.44	99.34 ± 0.79	98.56±0.42	99.24±0.45	**99.45 ± 0.41**	82.01±2.77	75.93 ± 2.28	78.36 ± 3.91	77.53±3.44	80.93±3.40	**76.21 ±2.91**
K× 100	85.20±1.31	97.72±1.26	99.33 ± 0.87	98.40±0.49	99.21±0.56	**99.42 ± 0.48**	66.76±6.35	67.09±5.89	67.15 ± 4.87	68.51±4.87	67.56±5.88	**68.90 ±7.01**

obtained results with the sampling rate of 5% and 10%. Similar to the results reported in Tables VI and VII, it is apparent that the HT-CNN-Attention consistently yields higher OA than that of other methods (i.e., EMP-SVM and CNN) on all the training samples. The proposed methods also have better performance with the controlled random sampling strategy.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

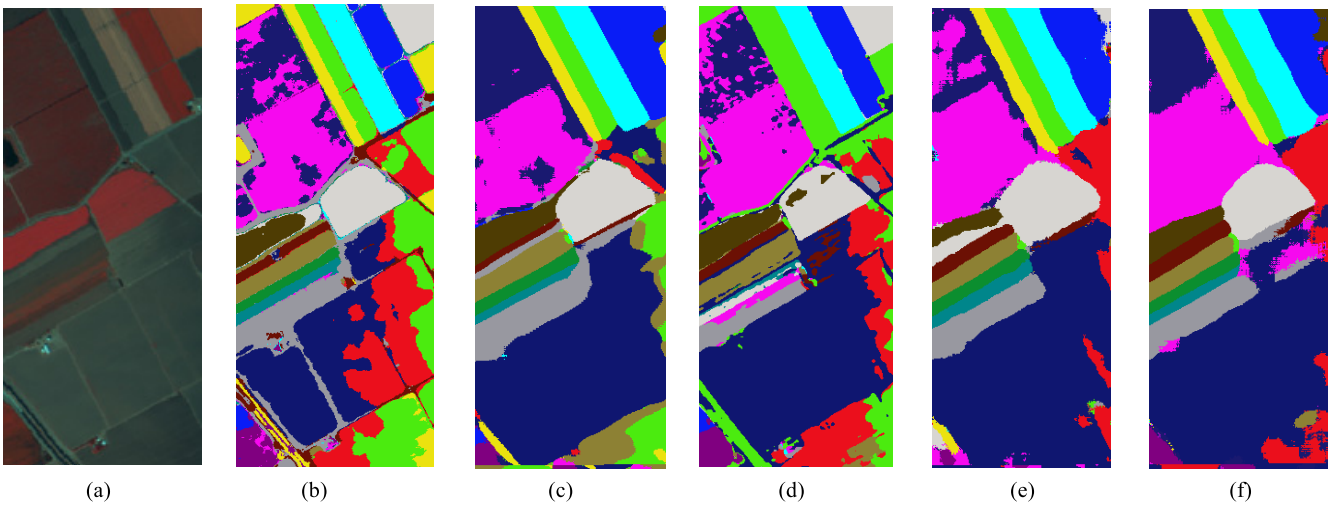


Fig. 15. Salinas. (a) False-color image. Classification maps of different classifiers (b) EMP-SVM, (c) CNN, (d) SSRN, (e) HT-CNN, and (f) HT-CNN-Attention.

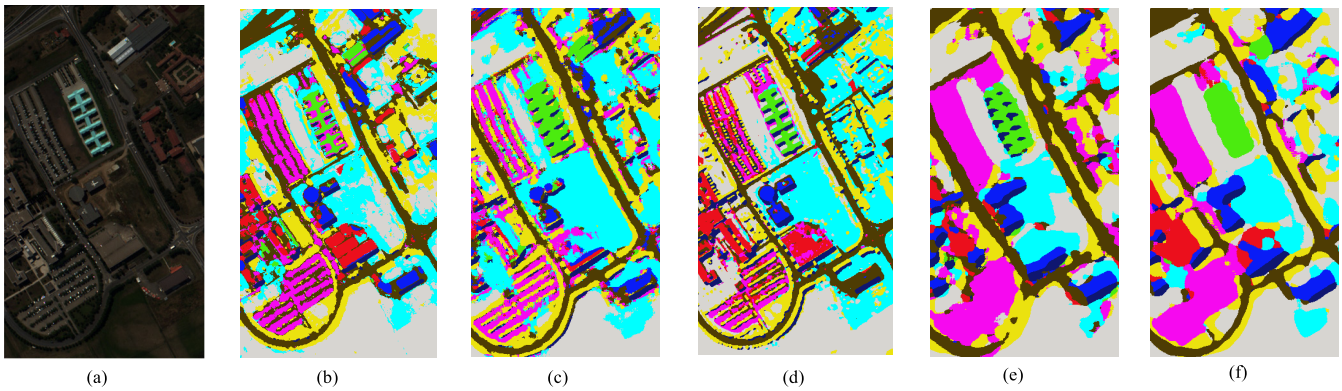


Fig. 16. Pavia. (a) False-color image. Classification maps of different classifiers (b) EMP-SVM, (c) CNN, (d) SSRN, (e) HT-CNN, and (f) HT-CNN-Attention.

Overall, based on all the abovementioned experiments, the proposed HT-CNN and HT-Attention are found effective to handle the task of HSI classification with different sampling strategies.

G. Discussion of Running Time

The processing time of different methods is reported in Tables I–V for both training and test stages. Here, we only consider 200 training samples. We conduct our experiments on a computer with an Intel Core i5-7300H processor with 2.5 GHz, 8 GB of DDR4 RAM, an NVIDIA GeForce GTX 1050Ti graphical processing unit (GPU).

As can be seen in Tables I–V, EMP-SVM demonstrates superior computational efficiency compared to other studied approaches, but the OA of EMP-SVM is low. Deep-learning-based methods only need a few minutes with 200 training samples using a GPU. SSRN costs more training time, because the number of training epoch is 200, and other deep-learning-based methods only demand dozens of training epochs. Compared with CNN, VGG consumes a longer training time, since it is composed of many 3×3 convolutional kernels and, therefore, has more parameters to train. For the T-CNN and RGB-T-CNN, the running time is shorter, because only three bands are used for the classification task. Compared to CNN and VGG, our proposed HT-CNN consumes less training time and reaches the highest accuracy, which demonstrates that our proposed method is an effective strategy.

Compared to HT-CNN, HT-CNN-Attention leads to the highest classification accuracy with an extremely small increase in computational complexity. For example, the training time increases by 4.36, 2.16, 9.49, and 3.8 s on the Salinas, Pavia, KSC, and Indian Pines data set, respectively.

H. Classification Map

The classification maps of different methods are presented in Figs. 15–18. The methods, including EMP-SVM, CNN, SSRN, HT-CNN, and HT-CNN-Attention, are selected to classify the whole images for four data sets. All parameters in these models are optimized. As can be seen, EMP-SVM always produces more errors compared to other studied methods for all four data sets. Especially, for the Salinas data set, many pixels are misclassified in the middle of this classification map with the caramel color. Compared to EMP-SVM, classification results of deep learning models, such as CNN and SSRN, contain fewer noisy scatter points in the classification maps. The proposed methods, which include

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

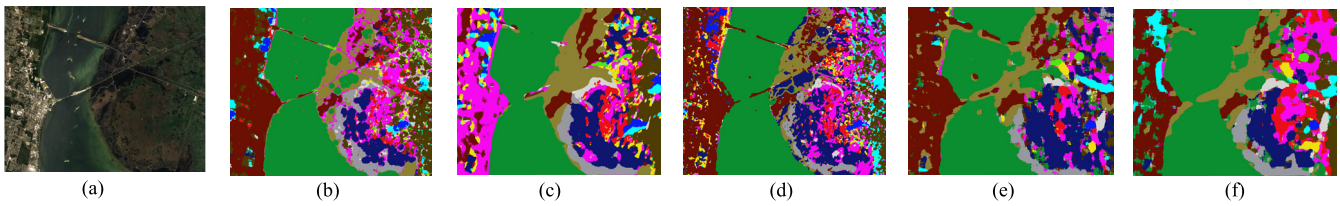


Fig. 17. KSC. (a) False-color image. Classification maps of different classifiers (b) EMP-SVM, (c) CNN, (d) SSRN, (e) HT-CNN, and (f) HT-CNN-Attention.

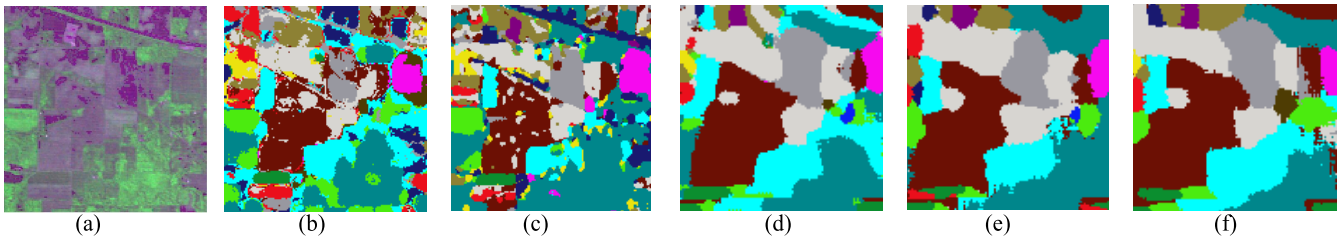


Fig. 18. Indian Pines. (a) False-color image. Classification maps of different classifiers (b) EMP-SVM, (c) CNN, (d) SSRN, (e) HT-CNN, and (f) HT-CNN-Attention.

HT-CNN and HT-CNN-Attention, are more precise for all four data sets. Our proposed methods with transfer learning work pretty well under the condition of the limited number of training samples, which demonstrates that the joint consideration of the CNN and transfer learning is promising for HSI classification.

V. Conclusion

In this article, a new classification framework entitled HT-CNN, which is the combination of transfer learning and deep CNN, has been proposed for HSI classification.

In the proposed method, the knowledge of source target, which is obtained by pretrained VGGNet-16, was transferred to the HSI classification model. The classification results have shown that the first several layers of deep CNN can extract general features. Therefore, the transfer learning strategy has potential for the HSI classification task, especially when the training samples are limited.

Channel mapping between the heterogeneous data sets has been proposed to handle the problem of different numbers of channels of ImageNet and HSI. The mapping layer is combined with the subsequent layers, and it can automatically map the inputs of HSI. The proposed mapping operation significantly improved the classification performance. Furthermore, compared with the original VGGNet-16 and other state-of-the-art deep learning methods (i.e., CNN and SSRN), the proposed methods significantly reduced the training time.

Due to the availability of huge differences between ImageNet and HSI, the obtained feature maps from the two data sets may also be different. Here, the attention mechanism has been introduced to adjust the importance of different feature maps and enhance classification accuracies.

The superiority of our proposed methods has been illustrated on four data sets with two training sample selection strategies (i.e., random sampling and controlled random sampling) by achieving better results in terms of classification accuracies. The HSI classification task with heterogeneous transfer learning is an effective approach independent of the choice of the sample selection strategies.

References

[1] C.-I Chang, *Hyperspectral Data Exploitation: Theory and Applications*. Hoboken, NJ, USA: Wiley, 2007.

[2] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. M. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[3] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[4] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Hyperspectral image classification using dictionary-based sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 10, pp. 3973–3985, Oct. 2011.

[5] P. Ghamisi *et al.*, "Advances in hyperspectral image and signal processing: A comprehensive overview of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 37–78, Dec. 2017.

[6] P. Ghamisi, J. Plaza, Y. Chen, J. Li, and A. J. Plaza, "Advanced spectral classifiers for hyperspectral images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 1, pp. 8–32, Mar. 2017.

[7] L. He, J. Li, C. Liu, and S. Li, "Recent advances on spectral-spatial hyperspectral image classification: An overview and new guidelines," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1579–1597, Mar. 2017.

[8] C.-I. Chang, Q. Du, T.-L. Sun, and M. L. G. Althouse, "A joint band prioritization and band-decorrelation approach to band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 6, pp. 2631–2641, Nov. 1999.

[9] B. Guo, S. R. Gunn, R. I. Damper, and J. D. B. Nelson, "Band selection for hyperspectral image classification using mutual information," *IEEE Geosci. Remote Sens. Lett.*, vol. 3, no. 4, pp. 522–526, Oct. 2006.

[10] Q. Wang, J. Lin, and Y. Yuan, "Salient band selection for hyperspectral image classification via manifold ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 6, pp. 1279–1289, Jun. 2016.

[11] G. Licciardi, P. R. Marpu, J. Chanussot, and J. A. Benediktsson, "Linear versus nonlinear PCA for the classification of hyperspectral data based on the extended morphological profiles," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 3, pp. 447–451, May 2011.

[12] T. M. Tu, "Unsupervised signature extraction and separation in hyperspectral images: A noise-adjusted fast independent component analysis approach," *Opt. Eng.*, vol. 39, no. 4, pp. 897–906, Apr. 2000.

[13] T. V. Bandos, L. Bruzzone, and G. Camps-Valls, "Classification of hyperspectral images with regularized linear discriminant analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 3, pp. 862–873, Mar. 2009.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

[14] T. Cover and P. Hart, "Nearest neighbor pattern classification," *IEEE Trans. Inf. Theory*, vol. IT-13, no. 1, pp. 21–27, Jan. 1967.

[15] J. Ledolter, "Multinomial logistic regression," in *Data Mining and Business Analytics With R*. Hoboken, NJ, USA: Wiley, 2013, pp. 109–124.

[16] S. Delalieux, B. Somers, B. Haest, T. Spanhove, J. V. Borre, and C. A. Mücher, "Heathland conservation status mapping through integration of hyperspectral mixture analysis and decision tree classifiers," *Remote Sens. Environ.*, vol. 126, pp. 222–231, Nov. 2012.

[17] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.

[18] S. Kawaguchi and R. Nishii, "Hyperspectral image classification by bootstrap AdaBoost with random decision stumps," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 11, pp. 3845–3851, Nov. 2007.

[19] M. Fauvel, J. A. Benediktsson, J. Chanussot, and J. R. Sveinsson, "Spectral and spatial classification of hyperspectral data using SVMs and morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3804–3814, Nov. 2008.

[20] L. Fang, N. He, S. Li, P. Ghamisi, and J. A. Benediktsson, "Extinction profiles fusion for hyperspectral images classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1803–1815, Mar. 2018.

[21] X. Kang, S. Li, and J. A. Benediktsson, "Spectral-spatial hyperspectral image classification with edge-preserving filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2666–2677, May 2014.

[22] Z. Wu *et al.*, "GPU parallel implementation of spatially adaptive hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1131–1143, Apr. 2018.

[23] L. Sun, Z. Wu, J. Liu, L. Xiao, and Z. Wei, "Supervised spectral–spatial hyperspectral image classification with weighted Markov random fields," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1490–1503, Mar. 2015.

[24] L. Fang, S. Li, W. Duan, J. Ren, and J. A. Benediktsson, "Classification of hyperspectral images by exploiting spectral-spatial information of superpixel via multiple Kernels," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 12, pp. 6663–6674, Dec. 2015.

[25] Y. Gu, J. Chanussot, X. Jia, and J. A. Benediktsson, "Multiple Kernel learning for hyperspectral image classification: A review," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6547–6565, Nov. 2017.

[26] J. Liu, Z. Wu, J. Li, A. Plaza, and Y. Yuan, "Probabilistic-kernel collaborative representation for spatial–spectral hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2371–2384, Apr. 2016.

[27] J. Li, P. R. Marpu, A. Plaza, J. M. Bioucas-Dias, and J. A. Benediktsson, "Generalized composite kernel framework for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 9, pp. 4816–4829, Sep. 2013.

[28] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[29] Y. Chen, X. Zhao, and X. Jia, "Spectral-spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[30] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. 25th Int. Conf. Neural Inf. Process. Syst.*, 2013, pp. 1097–1105.

[31] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[32] J. Yue, W. Zhao, S. Mao, and H. Liu, "Spectral-spatial classification of hyperspectral images using deep convolutional neural networks," *Remote Sens. Lett.*, vol. 6, no. 6, pp. 468–477, May 2015.

[33] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[34] E. Aptoula, M. C. Ozdemir, and B. Yanikoglu, "Deep learning with attribute profiles for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 12, pp. 1970–1974, Dec. 2016.

[35] Y. Chen, L. Zhu, P. Ghamisi, X. Jia, G. Li, and L. Tang, "Hyperspectral images classification with Gabor filtering and convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2355–2359, Dec. 2017.

[36] W. Li, G. Wu, F. Zhang, and Q. Du, "Hyperspectral image classification using deep pixel-pair features," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 844–853, Feb. 2017.

[37] W. Shao and S. Du, "Spectral-spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Oct. 2016.

[38] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[39] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "A new deep convolutional neural network for fast hyperspectral image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 120–147, Nov. 2018.

[40] S. Yu, S. Jia, and C. Xu, "Convolutional neural networks for hyperspectral image classification," *Neurocomputing*, vol. 219, pp. 88–98, Jan. 2017. [Online]. Available: http://linkinghub.elsevier.com/retrieve/pii/S0925231216310104

[41] S. Li, Q. Hao, G. Gao, and X. Kang, "The Effect of Ground Truth on Performance Evaluation of Hyperspectral Image Classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 12, pp. 7195–7206, Dec. 2018.

[42] J. Acquarelli, E. Marchiori, L. M. C. Buydens, T. Tran, and T. van Laarhoven, "Spectral-spatial classification of hyperspectral images: Three tricks and a new learning setting," *Remote Sens.*, vol. 10, no. 7, p. 1156, Jul. 2018.

[43] L. Mou, P. Ghamisi, and X. X. Zhu, "Unsupervised spectral–spatial feature learning via deep residual Conv–Deconv network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 391–406, Jan. 2018.

[44] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep&dense convolutional neural network for hyperspectral image classification," *Remote Sens.*, vol. 10, no. 9, p. 1454, Sep. 2018.

[45] J. Zhou, J. Liang, Y. Qian, Y. Gao, and L. Tong, "On the sampling strategies for evaluation of joint spectral-spatial information based classifiers," in *Proc. 7th WHISPERS*, Tokyo, Japan, Jun. 2015, pp. 1–4.

[46] J. Liang, J. Zhou, Y. Qian, L. Wen, X. Bai, and Y. Gao, "On the sampling strategy for evaluation of spectral-spatial methods in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 862–880, Feb. 2017.

[47] R. Hänsch, A. Ley, and O. Hellwich, "Correct and still wrong: The relationship between sampling strategies and the estimation of the generalization error," in *Proc. IGARSS*, Fort Worth, TX, USA, Jul. 2017, pp. 3672–3675.

[48] J. Lange, G. Cavallaro, M. Götz, E. Erlingsson, and M. Riedel, "The influence of sampling methods on pixel-wise hyperspectral image classification with 3D convolutional neural networks," in *Proc. IGARSS*, Valencia, Spain, Jul. 2018, pp. 2087–2090.

[49] M. Oquab, L. Bottou, I. Laptev, and J. Sivic, "Learning and transferring mid-level image representations using convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Columbus, OH, USA, Jun. 2014, pp. 1717–1724.

[50] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *Proc. ICLR*, San Diego, CA, USA, Apr. 2015, pp. 1–14.

[51] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. CVPR*, Miami, FL, USA, Jun. 2009, pp. 248–255.

[52] N. Kruger *et al.*, "Deep hierarchies in the primate visual cortex: What can we learn for computer vision?" *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 8, pp. 1847–1871, Aug. 2013.

[53] C. Szegedy *et al.*, "Going deeper with convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2015, pp. 1–9.

[54] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recog. (CVPR)*, Jun. 2016, pp. 770–778.

[55] G. Hinton, N. Srivastava, A. Krizhevsky, R. R. Salakhutdinov, and I. Sutskever, "Improving neural networks by preventing co-adaptation of feature detectors," *Comput. Sci.*, vol. 3, no. 4, pp. 212–223, Jul. 2012.

[56] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," in *Proc. 32nd Int. Conf. Mach. Learn.*, Feb. 2015, pp. 448–456.

[57] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowl. Data Eng.*, vol. 22, no. 10, pp. 1345–1359, Oct. 2010.

[58] J. Yosinski, J. Clune, Y. Bengio, and H. Lipson, "How transferable are features in deep neural networks?" in *Proc. Adv. Neural Inf. Process. Syst.*, Montreal, QC, Canada, Dec. 2014, pp. 3320–3328.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.

[59] S. Kornblith, M. Norouzi, H. Lee, and G. Hinton, "Similarity of neural network representations revisited," in *Proc. ICML*, Long Beach, CA, USA, May 2019, pp. 3519–3529.
[60] A. Gretton, O. Bousquet, A. Smola, and B. Schölkopf, "Measuring statistical dependence with Hilbert-Schmidt norms," in *Algorithmic Learning Theory*. Berlin, Germany: Springer, 2005, pp. 63–77.
[61] K. Xu *et al.*, "Show, attend and tell: Neural image caption generation with visual attention," in *Proc. ICML*, 2015, pp. 2048–2057.
[62] J. Cheng, L. Dong, and M. Lapata, "Long short-term memory-networks for machine reading," in *Proc. EMNLP*, Austin, TX, USA, Nov. 2016, pp. 551–561.
[63] Z. Lin *et al.*, "A structured self-attentive sentence embedding," in *Proc. ICLR*, Toulon, France, Mar. 2017, pp. 1–15.
[64] Y. S. Aurelio, G. M. de Almeida, C. L. de Castro, and A. P. Braga, "Learning from imbalanced data sets with weighted cross-entropy function," *Neural Process. Lett.*, vol. 50, no. 2, pp. 1937–1949, Oct. 2019.
[65] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.
[66] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional neural networks," in *Proc. 13th Eur. Conf. (ECCV)*, Colombia, SA, USA, Sep. 2014, pp. 818–833.



Xin He received the M.S. degree from the Harbin University of Science and Technology, Harbin, China. She is currently pursuing the Ph.D. degree with the School of Electronics and Information Engineering, Harbin Institute of Technology, Harbin.

Her research interest includes remote sensing image processing based on deep learning methods.



Yushi Chen received the Ph.D. degree from the Harbin Institute of Technology, Harbin, China, in 2008.

He is currently an Associate Professor with the School of Electronics and Information Engineering, Harbin Institute of Technology. His research interests include remote sensing data processing and machine learning.



Pedram Ghamisi (S'12–M'15–SM'18) is currently the Head of the Machine Learning Group, Helmholtz-Zentrum Dresden-Rossendorf, Dresden, Germany. His research interests include interdisciplinary research on remote sensing and machine (deep) learning, image, and signal processing, and multisensory data fusion.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 09:14:08 UTC from IEEE Xplore. Restrictions apply.